https://pubsonline.informs.org/journal/mnsc

MANAGEMENT SCIENCE
Vol. 69, No. 11, November 2023, pp. 6855–6875
ISSN 0025-1909 (print), ISSN 1526-5501 (online)

Scaling Up Electric-Vehicle Battery Swapping Services in Cities: A Joint Location and Repairable-Inventory Model

Wei Qi,[a] Yuli Zhang,[b,*] Ningwei Zhang[b]

[a] Department of Industrial Engineering, Tsinghua University, Beijing 100084, China; [b] School of Management and Economics, Beijing Institute of Technology, Beijing 100081, China

*Corresponding author

Contact: qiw@tsinghua.edu.cn, https://orcid.org/0000-0003-3948-835X (WQ); zhangyuli@bit.edu.cn, https://orcid.org/0000-0002-9382-6308 (YZ); zhangningwei@bit.edu.cn, https://orcid.org/0000-0003-3465-1322 (NZ)

Received: June 19, 2020
Revised: October 24, 2021; April 10, 2022; July 12, 2022
Accepted: September 29, 2022
Published Online in Articles in Advance: March 17, 2023

https://doi.org/10.1287/mnsc.2023.4731

Copyright: © 2023 INFORMS

Abstract. Battery swapping for electric vehicle refueling is reviving and thriving. Despite a captivating sustainable future where swapping batteries will be as convenient as refueling gas today, a tension is mounting in practice (beyond the traditional "range anxiety" issue): On one hand, it is desirable to maximize battery proximity and availability to customers. On the other hand, capacitated urban power grids may curb decentralized charging at a slow speed. To reconcile this tension, some cities are embracing an emerging infrastructure network: Decentralized swapping stations replenish charged batteries from centralized charging stations. It remains unclear how to design such a network or whether pooling charging demands will save costs or batteries. In this paper, we model this new urban infrastructure network. This task is complicated by non-Poisson swaps and by the intertwined stochastic operations of swapping, charging, stocking, and circulating batteries among swapping and charging stations. We tackle these complexities by deriving analytical models, which enrich the classical batched repairable-inventory theory. We next propose a joint location and repairable-inventory model for citywide deployment of hub charging stations, with a nonconvex nonconcave objective function. We solve this problem exactly by exploiting submodularity and combining constraint-generation and parameter-search techniques. Even for solving convexified problems, our algorithm brings a speedup of at least three orders of magnitude relative to the Gurobi solver. The major insight is twofold: The benefit of pooling charging demands alone is not enough to justify the adoption of the "swap-locally, charge-centrally" network; instead, the main justification is that faster charging accessible at centralized charging stations significantly reduces the system-wide battery stock level. In a broader sense, this work deepens our understanding of how mobility and energy are coupled toward enabling smart cities.

History: Accepted by Chung Piaw Teo, optimization.

Funding: Y. Zhang acknowledges the support from the National Natural Science Foundation of China [Grants 71871023, 72271029, and 72061127001]. W. Qi acknowledges the support from the National Natural Science Foundation of China [Grants 72272014 and 72188101] and the Natural Sciences and Engineering Research Council of Canada [Grant RGPIN-2019-04769]. N. Zhang acknowledges the support from the China Scholarship Council [202106030140].

Supplemental Material: The data files and online appendices are available at https://doi.org/10.1287/mnsc.2023.4731.

Keywords: battery swapping • electric vehicles • non-convex non-concave optimization algorithms • smart and sustainable cities

1. Introduction

Electrification of urban transportation is at the heart of the burgeoning smart-city transformations. Running on batteries and electric motors, electric vehicles (EVs) hold promises such as reducing carbon emissions (of which the transportation sector accounts for 24% globally (IEA 2020)), supporting electric power grids, and facilitating the transition into shared and autonomous mobility paradigms (Greenblatt and Saxena 2015). These anticipated benefits have pushed governments around the world to aggressively incentivize the adoption of EVs. For example, the United Kingdom and France plan to ban the sale of new internal-combustion vehicles by 2035 and 2040, respectively (NBC 2020). Chinese megacities ration more EV sales than fossil-fuel car sales by capping the numbers of issued license plates (Bloomberg 2019). Given these trends, more than 500 million passenger vehicles on the road, or 30% of the entire fleet, are expected to be electric by 2040 (BNEF 2019).

Over the last decade, the adoption of battery swapping grew less smoothly than that of EVs. Being an alternative to plugging in cars for battery charging, battery swapping refers to the mode of refueling an EV

by replacing the depleted battery with a charged one. Earlier companies pioneering battery swapping were struck with a major *phase I challenge*: The heavy upfront investment in building swapping stations and battery inventories was hardly financially viable when few customers used the service. One famous failure was the Israeli startup Better Place, which spent $850 million constructing a network of swapping stations before declaring bankruptcy in 2013.

As EVs thrive, battery swapping revives. For example, NIO, a Shanghai-based EV startup, once offered its car owners by 2020 lifelong battery swapping service for free and is now partnering with Sinopec to open 5,000 swapping stations by 2025 (Electrive 2021). The revival of battery swapping is driven by its three advantages over plug-in charging: (1) swiftness, a swap takes only three to five minutes, whereas using even Telsa's supercharger takes more than 30 minutes to charge a battery to 80%; (2) compactness, the short service time allows a swapping station to occupy much less space than a plug-in charging station to achieve the same service level; (3) safety, compared with vehicle owners, service providers can more properly charge and maintain batteries. Therefore, battery swapping is widely believed to prevail, especially in large cities.

Nevertheless, EV companies and municipalities are now facing major *phase II challenges* when they try to scale up battery swapping services in cities:

• *Service Proximity*: Whereas the phase I placement of swapping stations was mainly to assure drivers that they could reach a station within a driving range, the phase II deployment needs to be denser so that drivers can access a swapping station without many detours. The resulting urban swapping station networks are highly decentralized and thus more difficult to operate.

• *Battery Availability*: Swapping stations also need to ensure the high availability of charged batteries to satisfy swapping demands. This challenge is exacerbated by the decentralized layout of swapping stations because swapping demands are more variable when disaggregated than when pooled. Consequently, the service provider has to build up massive expensive battery inventories.

• *Grid Accessibility*: Finally, fast charging depleted batteries at swapping stations may overload or even destabilize local low-voltage power distribution grids. These "last-mile" grids were often built without allocating enough capacity for EV charging, as Muratori (2018) in *Nature Energy* identifies. As a result, only slow charging may be available. Upgrading existing distribution grids to accommodate fast charging at scale would be prohibitive, if not infeasible.

This paper aims to help city governments and practitioners address these phase II challenges by providing a deeper understanding of how to scale up citywide battery swapping services. In particular, the paper examines a *swap-locally, charge-centrally* network setting where batteries are locally swapped at decentralized swapping stations and transported to and charged at more centralized charging stations that are connected to grids of sufficient capacity (which are typically of a higher voltage level or near a substation). Such a setting was largely overlooked in the literature but becomes increasingly relevant in the presence of large adoption of battery swapping, possibly being the inevitable option to avoid overloading local power grids. In fact, one of the key strategies of the State Grid Corporation of China in developing China's EV infrastructure is to provide decentralized battery swapping as the primary refueling option with concentrated charging. Following this roadmap, cities such as Hangzhou have built large charging stations across the city as hubs to distribute charged batteries to local swapping stations (Xu et al. 2015). Nevertheless, how to deploy and operate such a network? Can pooling charge demands reduce the inventory of batteries and the system's cost? These questions remain unclear.

Our paper proposes data analytics, models, solution algorithms, and insights for the design and management of such swap-locally, charge-centrally service infrastructure networks that are expected to span our cities. The main contributions of this paper are as follows:

(1) To the best of our knowledge, this work is the first attempt to examine from the lens of management science the phase II infrastructural challenges that confront our cities in scaling up battery swapping. To this end, we expand the scope of classical joint location-inventory models to a more complex setting. Namely, we propose a joint location and repairable-inventory model to capture battery operations and charging station deployment.

(2) Our analysis of battery circulation extends the literature of multiechelon repairable inventory with batch ordering on four aspects: First, we relax the classical setup so that the demand arrivals can be non-Poisson, the order quantities and transportation times can be heterogenous, and the batched order arrival process is more accurately approximated. Second, we relax the charging/repair time to be random: not only across different batches but also within a batch. Third, we consider alternative policies for dispatching trucks between facilities. Collectively, we present more nuanced structural properties of batched repairable inventory systems. Finally, whereas classical repairable-inventory theory is generally of computational nature, our analytical models enable an expanded literature scope to citywide planning of multiple depots.

(3) Our paper contributes to the optimization literature by developing a new algorithmic framework based on the constraint-generation and parameter-search (CG&PS) techniques. The proposed joint location and the repairable-inventory problem is neither convex nor concave and thus does not permit a convex mixed integer reformulation. To address this difficulty, we use CG to tackle an equivalent mixed 0-1 linear

master problem by exploiting the hidden submodularity and design a customized PS to solve the nonconvex nonconcave *evaluation subproblem* exactly. Our algorithm exhibits superior computational efficiency. It solves large problem instances exactly within seconds. Even for convexified problems, our algorithm brings a speedup of at least three orders of magnitude compared with Gurobi while boasting superior accuracy over both the state-of-art branch-and-price algorithm and a local-search heuristic. The framework expands the scope of the problems that can be solved by using submodularity. Similar problem structures may extensively arise in stochastic inventory-location models and queueing-location models that involve additional decisions, such as order quantities of retailers and local server capacity.

(4) Our numerical studies demonstrate a two-fold insight regarding the scalability of centralizing battery charging. First, the benefit of inventory pooling alone cannot justify the adoption of the "swap-locally, charge-centrally" network. All else being equal, centralizing battery charging will actually incur a higher battery stock level than the existing mode of charging batteries at decentralized swapping stations due to two unfavorable effects: order batching and transportation lead time. Second, however, the main justification for the "swap-locally, charge-centrally" network is instead the capacitated urban power distribution grid that curbs decentralized charging at slow speed. In that case, sticking to decentralized charging will result in a significantly higher battery stock level and total cost. Therefore, it is imperative for battery swapping service providers to consider the swap-locally, charge-centrally strategy to ensure access to the grid for fast charging.

(5) The numerical studies also identify design and operational flexibilities for battery swapping services with centralized charging. (i) Cutting the quantity of batteries per order can reduce the amount of batteries needed with only mild cost increases. (ii) Adjusting the number of charging stations in a wide range around the optimal value does not significantly increase the total cost either. The service providers can thus exploit these flexibilities to adjust their design and operations to accommodate to various economic/environmental/societal complexities.

The remainder of this paper is organized as follows: Section 2 reviews the related literature. Section 3 introduces the model setup and data. Section 4 analyzes battery inventory management. Section 5 presents the models and the solution algorithm for citywide deployment of the battery supply network. Section 6 discusses numerical results and managerial insights. Section 7 concludes the paper. Notation tables, proofs, and additional analyses are available in the online appendices.

2. Literature Review

Scaling up battery swapping operations is essential to the future of sustainable urban mobility but has drawn little attention in the literature of operations research and operations management. The majority of the literature concerning EVs assume plug-in charging as the only refueling option. For example, Wu et al. (2021) devise a policy for charging plug-in EVs under time-varying electricity costs. They find it economically and environmentally beneficial to optimally delay charging given completion times that customers can select.

However, models and insights from those papers cannot be directly applied to the context of battery swapping. Swapping decouples batteries from vehicles, thus entailing completely different infrastructure and business operations. The literature on battery swapping largely focused on addressing the phase I challenges. For example, Mak et al. (2013) develop models for the initial deployment of swapping stations to enable early customer adoption. By comparison with the plug-in-only mode, Avci et al. (2015) find that battery swapping may lead to higher adoption but not necessarily more environmental benefits. Different from those papers, the recent phenomenal revival of battery swapping motivates us to shift focus from adoption to reconciling the tensions among service proximity, battery availability, grid accessibility, and operational flexibilities. Addressing these emerging phase II issues becomes imperative for scaling up battery swapping services.

From a theoretical perspective, our paper contributes to two bodies of literature. (1) First, our analysis of battery circulation enriches the multiechelon repairable inventory theory by relaxing the conventional setting while providing more nuanced analytical structural properties. We will elaborate on our connection to this literature in Section 4.1. (2) Second, our proposed CG&PS algorithm contributes to the literature of facility location models with nonlinear costs, which naturally arise in the joint location-inventory problem (Shen et al. 2003), warehouse-retailer network design problem (Teo and Shu 2004), and transportation-inventory network design problem (Shu et al. 2005). Existing solution techniques for such models with concave/convex/nondecreasing facility costs include branch-and-price, branch-and-bound based on Lagrangian relaxation, branch-and-cut based on the extended polynomial inequality, benders decomposition, and constant ratio approximation (Shu 2010, Li et al. 2013). See Table 2 in Online Appendix A for a comparison with existing exact algorithms. Our proposed algorithm differs from existing ones by two salient features:

(i) By exploiting the hidden submodularity, we propose a new mixed 0-1 linear reformulation to tackle the nonconvex nonconcave problem that does not permit a conic quadratic mixed integer program (CQMIP) reformulation. This new *master problem* reformulation complements the existing set-covering/partitioning and CQMIP formulations and shares a similar spirit with the extended polymatroid method. However, unlike

existing submodularity-based algorithms that adopt the branch-and-cut approach and rely on equivalent CQMIP formulations (Atamtürk and Narayanan 2008, Atamtürk et al. 2012, Atamtürk and Gómez 2020), we propose a CG framework that allows us to encapsulate the nonconvex nonconcave structure into the *evaluation subproblem*.

(ii) Existing polynomial time algorithms only solve special *unconstrained submodular* minimization problems (Shen et al. 2003, Teo and Shu 2004, Shu et al. 2005, Shen and Qi 2007, Snyder et al. 2007) or convex minimization problems (Fischetti et al. 2017) and thus do not apply to our evaluation subproblem. Compared with existing PS algorithms that solve *constrained discrete concave minimization/quasi-convex maximization* problems (Zhang et al. 2016a, b, 2018), the proposed customized PS algorithm is the first to exactly solve a *constrained continuous nonconvex nonconcave* problem. The algorithm turns out to be remarkably efficient by exploiting the separable structure of the *parameterized subproblem* (which permits closed-form solutions) and the monotonicity of the critical parameter over a series of evaluation subproblems.

More broadly, addressing the phase II challenges of battery swapping urges citywide planning and operations. In this sense, our paper is related to the emerging literature on smart-city operations. Problems arising in a smart-city context often blur the boundaries between sectors (e.g., mobility, logistics, and energy in our setting) and reflect multidimensional needs (e.g., efficiency and sustainability). See Hasija et al. (2020), Mak (2022), and Qi and Shen (2019) for more discussion.

Finally, we observe that the "swap-locally, charge-centrally" setting has been considered by several papers in the field of power systems engineering. Zhang and Wang (2016) propose a metaheuristic algorithm to dispatch batteries between swapping and charging stations. Liu et al. (2019) propose a deterministic mixed-integer linear program to schedule battery logistics and inventory while providing ancillary services to power systems. In comparison, our models preserve tractability while taking into account demand uncertainty and networked operations. In this sense, our models can also be potentially integrated into more comprehensive models in power systems studies.

3. Model Setup and Data

3.1. Topology

Consider battery swapping service networks that span a city as Figure 1(a) depicts. Each network consists of multiple local swapping stations and a central charging station. The central charging stations must be colocated with nodes of the power distribution grid with sufficient capacity (often with 10 kV or above voltage level). Local swapping stations face random EV arrivals each bringing a depleted battery in exchange for a charged one in stock.

3.2. Objective and Decisions

The primary objective of the battery swapping service provider is to minimize the sum of the infrastructure cost and the operational cost. The service provider also tries to minimize the number of required batteries, which are expensive. Whereas the phase I focus was on siting swapping stations to induce adoption, in phase II, swapping stations are exogenously deployed (often colocated with gas stations) to achieve a prespecified level of proximity to customers. Instead, the service provider needs to decide on the optimal locations of charging stations to serve the swapping stations. In doing so, the service provider also needs to optimize the operations of battery circulation.

3.3. Data

To realistically characterize the stochastic process of demand arrivals, we obtain access to the data repository of the National Monitoring and Management Center for New Energy Vehicles (NMMC-NEV) in China. NMMC-NEV monitors the real-time status of EVs in China, updated every five seconds. We examine the records in Beijing from December 1 to December 31, 2019. Each record includes a vehicle's unique ID number, time-stamped battery state of charge (SOC), and location. Our data suggest that, contrary to what the majority of literature assume, battery swap demand arrivals do not necessarily constitute a Poisson process. For more details, please see Online Appendix B.

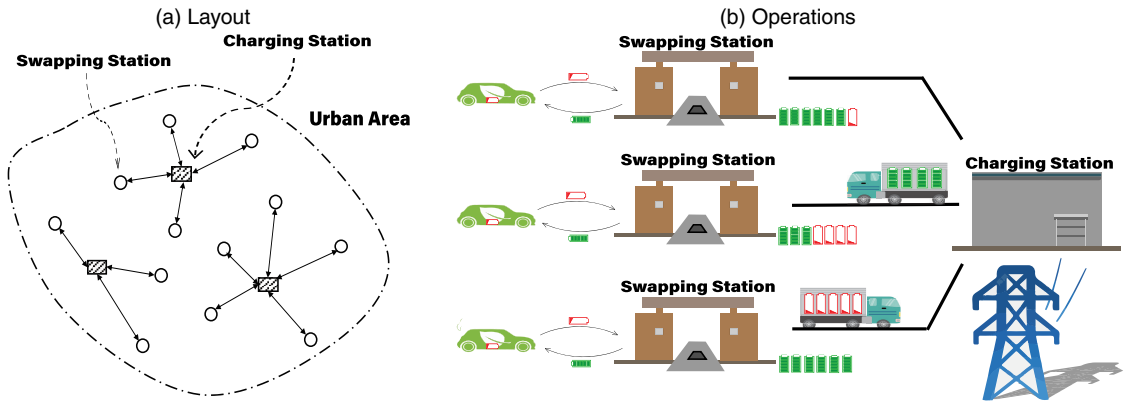
The data and statistical tests motivate us to model the unit-sized battery swap demand arrivals as a renewal process (Sivazlian 1974, Perera et al. 2018) with given mean and variance of the arrival rate. Together, the previous setup forms the basis for modeling the operations and design of citywide battery swapping service networks, which the following two sections elaborate on.

4. "Swap-Locally, Charge-Centrally" Operations

We start by modeling the operations of one service network, with one charging station located at $h \in \mathcal{H}$ serving a subset of swapping stations $\mathcal{I}_h$. Here $\mathcal{H}$ denotes the set of all feasible charging station locations; $\mathcal{I} \supseteq \mathcal{I}_h$ denotes the set of all swapping stations.

In particular, we examine the swapping, charging, stocking and circulation of batteries, as Figure 1(b) illustrates. Each swapping station adopts an (r, Q) policy to replenish its inventory of charged batteries. That is, once the stock level drops to a reorder level r_i, the swapping station operator at location i (with $r_i + Q_i$ charged batteries at time 0 in stock) will dispatch a truck to send the Q_i depleted batteries to the central charging station. In the meantime, the swapping station operator also requests Q_i batteries from the charging station. Upon this request, the charging station will deliver Q_i charged batteries to the swapping station i if sufficient inventory

Figure 1. (Color online) Swap-Locally, Charge-Centrally Battery Service Networks



is on hand; otherwise, a backorder occurs. We assume the amount of charged batteries that the charging station initially stocks is R. The goal of this section is to determine R and r_i for a given reorder quantity Q_i. Q_i will be jointly optimized in the infrastructure design problem in the next section.

We use $\{N_i^S(t); t \geq 0\}$ to denote the renewal process that counts the EV arrivals at a swapping station $i \in \mathcal{I}_h$ up to time t. The mean μ_i and variance σ_i^2 of the number of EV arrivals per hour are assumed to be known from the data. Each one-way truck trip takes time T_{ih}^T, and a full charge takes T^C. We assume both T_{ih}^T and T^C to be deterministic in Section 4.1 and relax this assumption in Section 4.2. Partially charged batteries are not allowed to swap to customers. For brevity, location indices are suppressed from symbols wherever the discussion holds for all locations.

4.1. Operations of a Charging Station

We first determine the charging station's stock level R. To this end, we introduce $D_i(t)$ to denote the cumulative *battery deficit*, which refers to the difference between $n_i^E(t)$, the cumulative number of batteries that swapping station i has requested up to time t, and $n_i^I(t)$, the cumulative number of batteries that swapping station i has transported to the charging station and have been fully charged by t. Intuitively, $n_i^E(t)$ and $n_i^I(t)$ can be viewed from the charging station as the battery "expenditure" to and "income" from swapping station i, respectively.

To formally define $n_i^E(t)$ and $n_i^I(t)$, by any time t (≥ 0), let $s_i(t)$ be the swapping station's last reorder time, and $k_i(t)$ be the cumulative number of orders that the swapping station has placed. It follows that $N_i^S(s_i(t)) = k_i(t)Q_i$, $k_i(t) \in \mathbb{N}_+$. Therefore, we can express the battery "expenditure" by time t as $n_i^E(t) = k_i(t)Q_i = N_i^S(s_i(t))$. Moreover, let $\Delta_i = T_i^T + T^C$ denote the *effective charging time*. For a depleted battery that a driver brought to the swapping station i before a reorder, this effective charging time refers to the time elapsed from the moment of that reorder to the completion of charging. Then at t, all the batteries that customers swapped to swapping station i before time $s_i(t - \Delta_i)$ have been transported to the charging station and received a full charge. Hence, the battery "income" by t is given by $n_i^I(t) = N_i^S(s_i(t - \Delta_i))$. The previous construction of the battery deficit $D_i(t) = n_i^E(t) - n_i^I(t)$ leads to three observations: (i) $\lim_{\Delta_i \to 0} D_i(t) = 0$ for any t; (ii) $D_i(t) \geq 0$ for any t; and (iii) every time $D_i(t) = 0$ for all $i \in \mathcal{I}_h$, the charging station resumes having R fully charged batteries on hand.

The charging station pools and shares the battery inventory with all swapping stations in the network. To set up a service level requirement at the charging station, suppose that swapping station i placed a reorder right before time t, so that the aggregate battery deficit $\sum_{i \in \mathcal{I}_h} D_i(t)$ increments by Q_i. If $\sum_{i \in \mathcal{I}_h} D_i(t)$ is still no greater than R, then it means that right before t at least Q_i charged batteries were available. Subsequently, Q_i of them were shipped out immediately, and there remain $R - \sum_{i \in \mathcal{I}_h} D_i(t)$ charged batteries at the charging station at t. Otherwise, if $\sum_{i \in \mathcal{I}_h} D_i(t) > R$, then it must have been the case that fewer than Q_i charged batteries were available right before t, and a backorder thus occurs. In short, the charging station observes a backorder quantity $(\sum_{i \in \mathcal{I}_h} D_i(t) - R)^+$. Suppose that the service provider imposes that the proportion of reorders that faces a backorder be less than ϵ^C. Then in the steady state, the probability of stockout, that is, $\sum_{i \in \mathcal{I}_h} D_i(t) > R$, should be less than ϵ^C:

$$\lim_{t \to \infty} \text{Prob}\left(\sum_{i \in \mathcal{I}_h} D_i(t) > R\right) < \epsilon^C. \tag{1}$$

We assume the battery deficit $D_i(t)$ to be independent across swapping stations, based on the following rationale and mild assumptions. (i) By definition, $n_i^I(t)$, $n_i^E(t)$ and thus $D_i(t)$ are stochastic processes that are completely driven by the swap demand arrival process $N_i^S(t)$ and given parameters $\{r_i, Q_i, T_i^T, T^C\}$. (ii) We assume that $N_i^S(t)$ faced by different swapping stations are

exogenous and independent from each other, given that the service levels at the charging station $(1-\epsilon^C)$ and at swapping stations are exogenously specified and guaranteed. (iii) Each swapping station independently follows an (r, Q) policy. Consequently, it suffices to analyze the battery deficit $D(t) = n^E(t) - n^I(t)$ incurred by one swapping station.

For brevity of the remaining exposition, we introduce three symbols:

$$X(t) \triangleq \frac{N^S(s(t)) - N^S(s(t-\Delta))}{Q},$$
$$Z(t) \triangleq N^S(t) - N^S(t-\Delta),$$
$$\hat{Y}(t) = \mathrm{mod}_Q(Z(t)),$$

where $\mathrm{mod}_A(B)$ denotes the remainder of dividing B by A. Intuitively, $X(t) \in \mathbb{Z}$ is the number of batches of depleted batteries that the swapping station has shipped after $s(t-\Delta)$; $Z(t) \in \mathbb{Z}$ is the number of swap demand arrivals since an effective charging time ago; and $\hat{Y}(t)$ is the remainder of dividing $Z(t)$ by Q. Moreover, when omitting the index t, we indicate these random variables in the steady state. For example, $\mathbb{E}[Z] = \mu\Delta$ and $\mathrm{Var}(Z) = \sigma^2\Delta$ are the mean and variance of $Z(t)$ in the steady state. See Figure 2 for illustration of the notation. The battery deficit is given by

$$D(t) = n^E(t) - n^I(t) = QX(t), \tag{2}$$

which is the number of depleted batteries shipped since $s(t-\Delta)$. The steady-state distribution of $D(t)$ is difficult to analyze directly. The main obstacle is that $D(t)$ is in batches and does not have a tractable form of the probability distribution function. However, based on Identity (2), we next show that $\mathbb{E}[D]$ has an expression, and $\mathrm{Var}(D)$ permits a tight approximation, both in compact forms.

4.1.1. Mean of Battery Deficit. The following result shows that the mean battery deficit increases linearly in Δ. In other words, reducing either T^T or T^C has a constant marginal effect on cutting the battery deficit on average. Conversely, $\mathbb{E}[D]$ is independent of the order quantity Q.

Result 1. *The mean of the battery deficit is* $\mathbb{E}[D] = \mathbb{E}[QX] = \mathbb{E}[Z] = \Delta\mu$.

4.1.2. Variance of Battery Deficit. To analyze $\mathrm{Var}(D)$, we first identify the following connection.

Lemma 1. *The variance of the battery deficit is given by* $\mathrm{Var}(D) = \mathrm{Var}(QX) = \mathrm{Var}(Z) + Q\mathbb{E}[\hat{Y}] - \mathbb{E}[\hat{Y}^2]$.

Lemma 1 shows that $\mathrm{Var}(QX)$ is greater than $\mathrm{Var}(Z)$ by a margin $Q\mathbb{E}[\hat{Y}] - \mathbb{E}[\hat{Y}^2]$, which is greater than zero when $Q > 1$. The insight here is that consolidating orders into batches of $Q > 1$ creates extra operational uncertainties compared with a continuous ordering process with $Q = 1$.

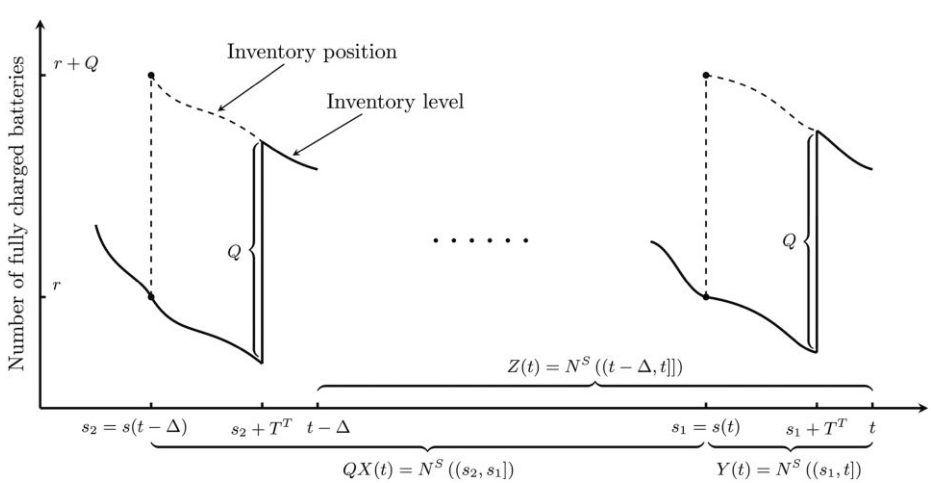
However, the expression in Lemma 1 is inconvenient for further analysis because $\mathbb{E}[\hat{Y}]$ and $\mathbb{E}[\hat{Y}^2]$ do not admit closed-form expressions. To overcome this difficulty, we find that $\mathrm{Var}(QX)$ exhibits different structural properties depending on the value of the order quantity Q relative to $\mathbb{E}[Z]$. For the case where Q is far less than $\mathbb{E}[Z]$, the following lemma gives the expression of $\mathrm{Var}(QX)$.

Lemma 2. *If the steady-state distribution of* $\hat{Y}(t)$ *is uniform over* $\{0, 1, \ldots, Q-1\}$, *then*

$$\mathrm{Var}(QX) = \mathrm{Var}(Z) + \frac{Q^2-1}{6} = \Delta\sigma^2 + \frac{Q^2-1}{6}. \tag{3}$$

The technical condition in Lemma 2 asymptotically holds when $\frac{Q}{\mathbb{E}[Z]}$ goes to zero. Intuitively, for small Q, the probability mass of $Z(t)$ in the steady state does not vary significantly over $\{lQ, lQ+1, \ldots, (l+1)Q-1\}$ for any integer l. As for the other extreme case where Q is

Figure 2. Inventory Position and Level of Charged Batteries in a Swapping Station Under the (r, Q) Policy



far greater than $\mathbb{E}[Z]$, the following lemma provides an alternative expression of $\mathrm{Var}(QX)$,

Lemma 3. *If* $\mathbb{P}(Z \geq Q) = 0$, *then*

$$\mathrm{Var}(QX) = Q\mathbb{E}[Z] - (\mathbb{E}[Z])^2 = Q\Delta\mu - (\Delta\mu)^2. \tag{4}$$

The condition $\mathbb{P}(Z \geq Q) = 0$ asymptotically holds when $\frac{Q}{\mathbb{E}[Z]}$ goes to infinity. Lemma 3 suggests that $\mathrm{Var}(QX)$ is linear in Q. Intuitively, under this condition, Q is so large (and thus the reorder frequency is so low) that at most one reorder occurs per effective charging time Δ. Consequently, $QX(t)$ can only take value of either zero, when $s(t) = s(t-\Delta)$, or Q, when $s(t) > s(t-\Delta)$. When Q increases, on one hand, the variation of QX amplifies quadratically; on the other hand, $\mathbb{P}(X = 0)$ also increases, which reduces the variation of QX.

In what follows, we use the following piecewise function to approximate $\mathrm{Var}(QX)$:

$$\phi(Q) = \begin{cases} \Delta\sigma^2 + \dfrac{Q^2 - 1}{6}, & \text{if } 0 \leq Q \leq \nu, \\ Q\Delta\mu - (\Delta\mu)^2, & \text{if } Q \geq \nu, \end{cases} \tag{5}$$

where $\nu = 3\Delta\mu - \sqrt{3(\Delta\mu)^2 - 6\Delta\sigma^2 + 1}$ is the first intersection point between the two pieces of the function. ν is a positive number. Thus, we assume $3(\Delta\mu)^2 - 6\Delta\sigma^2 + 1 \geq 0$. The approximation given by (5) turns out to be remarkably accurate for all Q, as the numerical example later in this section will show. We provide two rationales behind the efficacy of this approximation. First, Lemma 4 provides unified bounds of $\mathrm{Var}(X)$, and suggests that (3) in Lemma 2 is reasonably tight for small values of Q, and (4) in Lemma 3 serves as a lower bound for all Q.

Lemma 4. *The variance of QX is bounded from above and from below as follows:*

$$\max\{Q\mathbb{E}[Z] - \mathbb{E}[Z]^2, \mathrm{Var}(Z)\} \leq \mathrm{Var}(QX) \leq \mathrm{Var}(Z) + \frac{Q^2}{4}.$$

Second, even for a moderate value of Q (e.g., $\approx 1.3\mathbb{E}[Z]$), the probability that more than one reorder occur during an effective charging time is very small. Therefore, following the discussion after Lemma 3, we believe that (4) is satisfactorily tight for moderate or large values of Q. Collectively, the previous analysis leads to the following approximation of the variance of battery deficit $\mathrm{Var}(D)$.

Result 2. *The piecewise expression $\phi(Q)$ in (5) approximates* $\mathrm{Var}(D)$.

To further test the accuracy of this approximation, we numerically generate 0.5 million customer arrivals. Their interarrival times follow a normal distribution truncated at zero, so that $\mu = 8\mathrm{hr}^{-1}$ and $\sigma^2 = 9.4\mathrm{hr}^{-1}$. In addition, $T^T = 0.5$ hours, $T^C = 2$ hours, and $r = 8$. Figure 3 illustrates the simulated and the approximated standard deviation (as well as mean) of battery deficit as Q varies. It shows that the approximation error is negligibly small. We repeat this simulation with different parameter values and probability distributions to generate swap demand arrivals, consistently observing small approximation errors. Hence, in the following, we assume that $\mathrm{Var}(D)$ is given by $\phi(Q)$ in (5).

4.1.3. Battery Stock Level. Given the previous analysis, the aggregate battery deficit $\sum_{i\in\mathcal{I}_h} D_i(t)$ has mean $\sum_{i\in\mathcal{I}_h}\Delta_i\mu_i$ and variance $\sum_{i\in\mathcal{I}_h}\phi_i(Q_i)$ in the steady state. The steady-state distribution of $\sum_{i\in\mathcal{I}_h} D_i(t)$ can be approximated in multiple ways. We choose the normal distribution $\sum_{i\in\mathcal{I}_h} D_i \sim \mathrm{N}(\sum_{i\in\mathcal{I}_h}\Delta_i\mu_i, \sum_{i\in\mathcal{I}_h}\phi_i(Q_i))$ with three reasons. (i) The normal distribution is mathematically convenient. (ii) The normal distribution represents a risk-neural approximation and thus allows a more straightforward comparison across different scenarios. (iii) Following the central limit theorem, if $D_i(t)$ are mutually independent, $\sum_{i\in\mathcal{I}_h} D_i(t)$ will converge to a normal random variable as cardinality $|\mathcal{I}_h|$ increases (under mild conditions). We expect that the number of swapping stations served by one charging station will increase as the battery swapping service scales up.

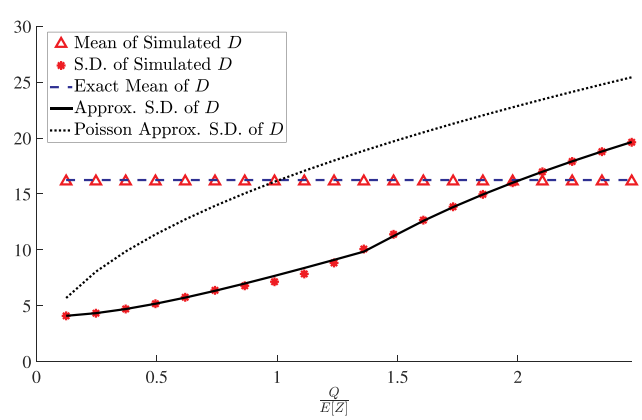
Subsequently, the service level requirement in (1) leads to the following result on the battery stock level at a central charging station.

Result 3. *The number of batteries to stock at a charging station h needs to satisfy*

$$R \geq \sum_{i\in\mathcal{I}_h}\Delta_i\mu_i + \Phi^{-1}(1-\epsilon^C)\sqrt{\sum_{i\in\mathcal{I}_h}\phi_i(Q_i)}, \tag{6}$$

where $\Phi^{-1}(\cdot)$ is the inverse of the cumulative standard normal distribution function.

Figure 3. (Color online) Simulated vs. Approximated Standard Deviation of Battery Deficit D



4.1.4. Connection and Contribution to the Repairable-Inventory Literature. Lemmas 1–4 are key technical results of the modeling part of this paper. These results enrich the repairable inventory theory (see Guide and Srivastava (1997) for a comprehensive review). In a conventional setting, the aggregate battery deficit $\sum_{i\in\mathcal{I}_h} D_i(t)$ is equivalent to the total number of items that are either being repaired at the depot (i.e., the charging station) or in transit from the bases (i.e., the swapping stations) to the depot, as "$Q_0(t)$" in Graves (1985) or "$r(t)Q$" in Lee and Moinzadeh (1987). We would approximate the variance of $\sum_{i\in\mathcal{I}_h} D_i(t)$ differently if we instead followed these classical repairable inventory models with batch ordering, which rely on the following assumptions/rationales (Lee and Moinzadeh 1987). (i) The arrival process of failed items at each base is Poisson with the rate μ_i. (ii) The order quantity Q^c (as well as the transportation time) is the same for all bases. (iii) The process of order arrivals at the depot is a superposition of $|\mathcal{I}_h|$ arrival processes each having an Erlang-distributed interarrival time. (iv) This aggregate process is then approximated by a Poisson process with rate $\sum_{i\in\mathcal{I}_h}\mu_i/Q^c$. (v) Finally, the depot can be viewed as an $M_{Q^c}/G/+\infty$ queueing system with Poisson batch arrivals of size Q^c. Because of Palm's theorem, the number of batches being sent to or repaired at the depot, that is, $\sum_{i\in\mathcal{I}_h} D_i(t)$, is Poisson distributed, with the mean being the ratio of $\sum_{i\in\mathcal{I}_h}\mu_i/Q^c$ to the common service rate. Clearly, following this logic would approximate $\mathrm{Var}(\sum_{i\in\mathcal{I}_h} D_i)$ as $\sum_{i\in\mathcal{I}_h}\mu_i\Delta Q^c$ (as opposed to $\sum_{i\in\mathcal{I}_h}\phi_i(Q_i)$).

To summarize, the conventional repairable-inventory models simply approximate $\sum_{i\in\mathcal{I}_h} D_i(t)$ as Poisson distributed and focus on providing exact computational procedures or analytic approximations for expected backorders at the depot and bases (Sherbrooke 1968, Simon 1971, Graves 1985, Moinzadeh and Lee 1986). However, our numerical study shows that this approximation deviates from the true value by a large margin in our setting, as Figure 3 shows.

To understand this discrepancy in approximating $\mathrm{Var}(\sum_{i\in\mathcal{I}_h} D_i)$, we relax the previous conventional setting in three ways. First, we allow both the order quantity and the transportation time to be swapping-station specific. Second, we use both the mean and the variance to characterize the demand process. Third, we avoid approximating the batched order arrival process with a Poisson process and provide more accurate characterization for $\sum_{i\in\mathcal{I}_h} D_i(t)$. Consequently, Lemmas 1–4 reveal more nuanced structural properties of inner workings of circulation of batteries or repairable items than classical models could, discerning different roles that the order quantity Q_i plays when it is relatively large or small. These results facilitate the infrastructure planning model.

For more discussion on additional complexities such as finite charging/repair capacity, multiple charging/repair depots, and alternative truck dispatch policies, please see Online Appendix E. In what follows, we relax the effective charging time to be random—not only across different batches but also within a batch. In contrast, classical models typically assume a random repair time, yet with the restriction that all units in one batch have the same repair time (Shanker 1981, Moinzadeh and Lee 1986). This restriction is unrealistic in the context of battery swapping.

4.2. Random Effective Charging Time

To capture the randomness of effective charging time, we again analyze the battery deficit $D(t)$ incurred by one swapping station. Now suppose that the effective charging time $\Delta = T^T + T^C$ is drawn from the following discrete distribution (which may vary by swapping station):

$$\mathrm{Prob}(\Delta = \Delta_l) = p_l, \quad \forall l \in \mathcal{L},$$

where $\mathcal{L}$ is a finite index set. We denote the queue at the charging station by $GI_Q/DR/+\infty$. Here GI_Q denotes the general bulk arrival process with batch size Q. DR indicates that all effective charging times, be they within a batch or across different batches, are i.i.d. random samples from the given discrete probability distribution.

4.2.1. Modeling Battery Deficit. As Figure 4 illustrates, our analysis is based on the following additional notation: The $GI_Q/DR/+\infty$ queue can be decomposed into $|\mathcal{L}|$ subqueues each denoted by $GI_Q^{U^l}/\Delta_l/+\infty$, where U^l is the random number of batteries in a batch that require a deterministic effective charging time of Δ_l. Clearly, U^l follows a binomial distribution $B(Q, p_l)$. Furthermore, define

$$X^l(t) \triangleq \frac{N^S(s(t)) - N^S(s(t-\Delta_l))}{Q},$$
$$Z^l(t) \triangleq N^S(t) - N^S(t-\Delta_l),$$
$$\hat{Y}^l(t) = \mathrm{mod}_Q(Z_l(t)),$$

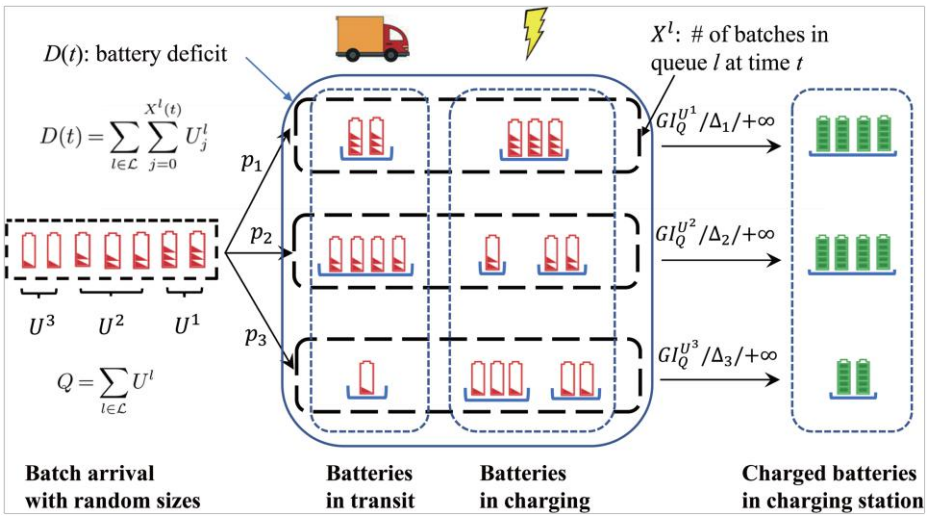
so that $X^l(t)$ is the random number of batches that the swapping station has shipped to (and reordered from) the charging station since time $s(t-\Delta_l)$. Our preceding analysis yields that

$$\mathbb{E}[X^l] = \frac{\mathbb{E}[Z^l]}{Q} = \frac{\mu\Delta_l}{Q}, \tag{7a}$$

$$\mathrm{Var}(X^l) = \frac{1}{Q^2}\left(\mathrm{Var}(Z^l) + Q\mathbb{E}\left[\hat{Y}^l\right] - \mathbb{E}\left[\left(\hat{Y}^l\right)^2\right]\right) \approx \frac{\phi_Q(\Delta_l)}{Q^2}, \tag{7b}$$

where $\phi_Q(\cdot)$ is given by the piecewise function (5) but now written explicitly as a function of effective charging time. We index those $X^l(t)$ batches (if any) chronologically as $\{k = 1, 2, \ldots\}$, and let U_k^l be the number of batteries in batch k with charging time Δ_l. Then $\{U_k^l:$

Figure 4. (Color online) Decomposing the $GI_Q/DR/+\infty$ Queue at the Charging Station into $|\mathcal{L}| = 3$ Subqueues (Each Denoted by $GI_Q^{U^l}/\Delta_l/+\infty$) in the Presence of Effective Charging Time Randomness



$k = 1, 2, \ldots\}$ are i.i.d., with $\mathbb{E}[U_k^l] = Qp_l$ and $\mathrm{Var}(U_k^l) = Qp_l(1-p_l)$; $U_0^l = 0$ when $X^l(t) = 0$.

Moreover, define $D^l(t)$ as the number of batteries that the swapping station has shipped since $s(t-\Delta_l)$ that require a charging time of Δ_l, that is, $D^l(t) = \sum_{j=0}^{X^l(t)} U_j^l$. Subsequently, the battery deficit is given by $D(t) = \sum_{l\in\mathcal{L}} D^l(t)$. Notice that $D(t)$ no longer necessarily equals $QX(t)$ but depends on random charging times. Also, $D(t)$ here can still be assumed to be independent across swapping stations under the mild assumptions of independent swap arrival processes and independent replenishment policies as in Section 4.1 because the decomposition into $|\mathcal{L}|$ subqueues has nothing to do with the stock levels of the charging station or any other swapping station.

4.2.2. Mean and Variance of Battery Deficit. We next outline the main steps of obtaining the mean and the variance of $D(t)$ in the steady state. We first examine $D^l(t)$ in the following lemma.

Lemma 5. *In the steady state for any $l \in \mathcal{L}$,*

(i) *the mean of $D^l(t)$ is $\mathbb{E}[D^l] = \mathbb{E}[X^l]\mathbb{E}[U^l] = \mu\Delta_l p_l$ and*

(ii) *the variance of $D^l(t)$ is $\mathrm{Var}(D^l) = \mathbb{E}[X^l]\mathrm{Var}(U^l) + \mathrm{Var}(X^l)\mathbb{E}[U^l]^2 \approx \mu\Delta_l p_l\,(1-p_l) + p_l^2\phi_Q(\Delta_l)$.*

Lemma 5(i) immediately leads to the following expression of $\mathbb{E}[D]$.

Result 4. *The mean battery deficit by a swapping station under charging time uncertainty is*

$$\mathbb{E}[D] = \mathbb{E}\left[\sum_{l\in\mathcal{L}} D^l\right] = \mu\sum_{l\in\mathcal{L}} p_l\Delta_l = \mu\mathbb{E}[\Delta]. \tag{8}$$

Result 4 is expected. It shows that our previous model with deterministic charging time can unbiasedly estimate $\mathbb{E}[D]$ if approximating the random Δ by its mean value. Lemma 5(ii), however, shows that $\mathrm{Var}(D^l)$ is greater than $\mathrm{Var}(p_l QX^l)$ (i.e., the second term, $\mathrm{Var}(X^l)\mathbb{E}[U^l]^2$). The extra variation, namely $\mathbb{E}[X^l]\ \mathrm{Var}(U^l)$, is contributed by the uncertainty of charging time.

Given Lemma 5(ii) and the following formula:

$$\mathrm{Var}(D) = \sum_{l\in\mathcal{L}} \mathrm{Var}(D^l) + \sum_{1\le l\ne s\le|\mathcal{L}|} (\mathbb{E}[D^l D^s] - \mathbb{E}[D^l]\mathbb{E}[D^s]), \tag{9}$$

it remains to derive the covariance $\mathbb{E}[D^l D^s] - \mathbb{E}[D^l]\mathbb{E}[D^s]$, which the following lemma does. Without loss of generality, suppose $\Delta_1 \le \Delta_2 \le \cdots \le \Delta_{|\mathcal{L}|}$ so that $X^l \le X^s$ holds for any $1 \le l < s \le |\mathcal{L}|$.

Lemma 6. *For any $l < s$, the covariance of $D^l(t)$ and $D^s(t)$ is given by*

$$\begin{aligned}\mathbb{E}[D^l D^s] - \mathbb{E}[D^l]\mathbb{E}[D^s] = {} & p_l p_s Q^2 \mathrm{Var}(X^l) + p_l p_s Q^2(\mathbb{E}[X^l(X^s - X^l)] \\ & - \mathbb{E}[X^l]\mathbb{E}[X^s - X^l]) - p_l p_s Q\mathbb{E}[X^l].\end{aligned} \tag{10}$$

On the right-hand side (RHS) of (10), only $\mathbb{E}[X^l(X^s - X^l)]$ does not admit a closed-form expression. Nevertheless, we identify the following properties depending on the order frequency/quantity.

Lemma 7. *For any $l < s$, $\mathbb{E}[X^l(X^s - X^l)]$ satisfies the following properties:*

(i) *If $\mathbb{P}(t - s(t) \le \Delta_s - \Delta_l) = 1$ for any t, then $\mathbb{E}[X^l(X^s - X^l)] = \mathbb{E}[X^l]\mathbb{E}[X^s - X^l] = \frac{\mu^2\Delta_l(\Delta_s - \Delta_l)}{Q^2}$.*

(ii) *If $\mathbb{P}(Z^s \ge Q) = 0$, then $\mathbb{E}[X^l(X^s - X^l)] = 0$.*

Intuitively, the technical condition in Lemma 7(i) holds when the order frequency is sufficiently high. Under this condition, $s(t-\Delta_s) < t-\Delta_s \le s(t-\Delta_l) < t-\Delta_l$, so that the two intervals $(s(t-\Delta_s), s(t-\Delta_l)]$ and $(s(t-\Delta_l), t-\Delta_l]$ are disjoint. Consequently, $(X^s - X^l)$, which counts the orders during $(s(t-\Delta_s), s(t-\Delta_l)]$, and X^l, which counts the orders during $(s(t-\Delta_l), t-\Delta_l]$, become independent. On the opposite end, the technical condition in Lemma 7(ii) has been used and interpreted in Lemma 3 to ensure extremely low order frequency and large order quantity. Under this condition, X^s and X^l can only be either one or zero, so that $X^l(X^s - X^l)$ is always zero.

Lemma 7 motivates us to approximate the covariance $Q^2(\mathbb{E}[X^l(X^s - X^l)] - \mathbb{E}[X^l]\mathbb{E}[X^s - X^l])$ by the following piecewise function:

$$\psi_Q(\Delta_l, \Delta_s) = \begin{cases} 0, & \text{if } 0 \le Q \le \nu, \\ -\mu^2 \Delta_l(\Delta_s - \Delta_l), & \text{if } Q \ge \nu, \end{cases} \tag{11}$$

which is in similar fashion as $\phi_Q(\Delta_l)$ approximates $\mathrm{Var}(QX^l)$. Collectively, substituting Lemma 5(ii), Lemma 6, and (11) into Equation (9) yields the following approximation of $\mathrm{Var}(D)$.

Result 5. *The variance of the battery deficit is approximated by*

$$\mathrm{Var}(D) \approx \mu \sum_{l\in\mathcal{L}} \Delta_l p_l (1 - p_l - 2\hat{p}_l) + \sum_{l\in\mathcal{L}} p_l(p_l + 2\hat{p}_l)\phi_Q(\Delta_l) + \sum_{1\le l\ne s\le|\mathcal{L}|} p_l p_s \psi_Q(\Delta_l, \Delta_s), \tag{12}$$

in which $\hat{p}_l \triangleq \sum_{s=l+1}^{|\mathcal{L}|} p_s$.

To further appreciate the approximation given by (12), it is interesting to compare it with the naïve sample-average approximation: $\mathrm{Var}(D) \approx \sum_{l\in\mathcal{L}} p_l \phi_Q(\Delta_l)$, which completely ignores the effective charging time variability within a batch. For this purpose, we establish the following result.

Proposition 1. *Any (discrete) effective charging time distribution has the following inequalities:*

(i) $\sum_{l\in\mathcal{L}} \Delta_l p_l (1 - p_l - 2\hat{p}_l) > 0$.

(ii) $\sum_{l\in\mathcal{L}} p_l(p_l + 2\hat{p}_l)\phi_Q(\Delta_l) < \sum_{l\in\mathcal{L}} p_l \phi_Q(\Delta_l)$.

Proposition 1 immediately suggests mixed effects of introducing nonidentical effective charging times within a batch: Compared with the sample-average approximation, on one hand, Proposition 1(i) shows that the first term of (12) increases $\mathrm{Var}(D)$. That is, allowing batteries in a batch to have nonidentical effective charging times contributes an extra source of battery deficit variation, which is proportional to the swap demand rate μ. On the other hand, the second term being less than $\sum_{l\in\mathcal{L}} p_l \phi_Q(\Delta_l)$ (from Proposition 1(ii)) and the third term being nonpositive (from (11)) together reduce $\mathrm{Var}(D)$, thus exhibiting a risk-pooling effect. This is because allowing effective charging times to be i.i.d. (as opposed to being sampled at the batch level) reduces the cross-batch variability of charging times.

We also numerically validate the proposed approximation expression (12) for the battery deficit variance. Figure 5 shows that (12) outperforms the naïve sample-average approximation, incurring smaller error. Hence, one can proceed to use (8) and (12) to construct the normal distribution of the aggregate battery deficit incurred by all swapping stations and then derive the battery stock level at the central charging station, in a similar fashion as in the previous section.

Result 6. *In the presence of random effective charging time, the number of batteries to stock at a charging station h needs to satisfy*

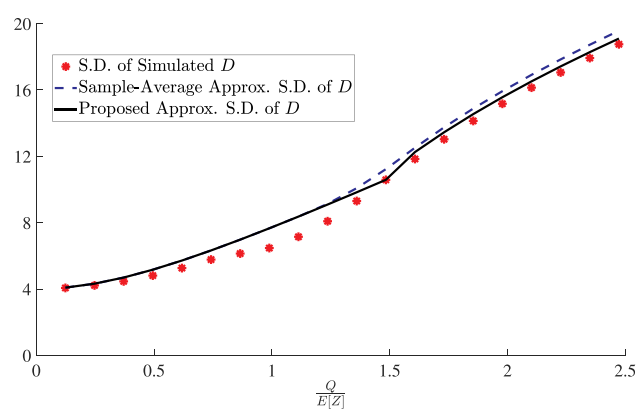
$$R \ge \sum_{i\in\mathcal{I}_h} \mu_i \sum_{l\in\mathcal{L}} p_{il}\Delta_{il} + \Phi^{-1}(1-\epsilon^C) \cdot \left(\sum_{i\in\mathcal{I}_h} \left(\mu_i \sum_{l\in\mathcal{L}} \Delta_{il} p_{il}(1 - p_{il} - 2\hat{p}_{il}) + \sum_{l\in\mathcal{L}} p_{il}(p_{il} + 2\hat{p}_{il})\phi_Q(\Delta_{il}) + \sum_{1\le l\ne s\le|\mathcal{L}|} p_{il}p_{is}\psi_Q(\Delta_{il}, \Delta_{is}) \right) \right)^{\frac{1}{2}}. \tag{13}$$

To summarize, this section further extends the classical repairable-inventory theory by accounting for nonidentical service times within each batch. Although the resulting models are more nuanced than those in the previous section, the model structures remain unchanged. Therefore, we relegate further model development into Online Appendix F and still assume deterministic transportation and charging time hereafter in the main text to keep the notation brief without compromising main managerial insights.

4.3. Operations of a Swapping Station

We next determine the reorder point r for each swapping station. A swapping station may run out of stock during the lead time between a reorder and the

Figure 5. (Color online) Proposed vs. Sample Average Approximation for the Standard Deviation of Battery Deficit D



subsequent truck delivery of Q batteries. This lead time consists of a fixed transportation time T^T and a random retard time, denoted by T^D, due to backordering at the charging station. Assume that the service provider imposes this stockout probability to be no greater than ϵ^S. Then the reorder level r must satisfy

$$\text{Prob}(N^S(T^T + T^D) \geq r + 1) \leq \epsilon^S. \tag{14}$$

The probability distribution of T^D is difficult to characterize because T^D intricately depends on decisions R, r_i, and Q_i for all $i \in \mathcal{I}_h$. Therefore, we propose a two-step procedure to obtain r. Moinzadeh and Lee (1986) also propose a two-step procedure to separately optimize Q and (r, R) numerically based on a power approximation of the total stock and backorders in the system.

In the first step, we ignore T^D. This simplification immediately leads to the following result if we approximate the probability distribution of $N^S(T^T)$ by a normal distribution matching the mean $T^T\mu$ and variance $T^T\sigma^2$ (normal distribution is widely used in literature to characterize road traffic, such as in Pak-Poy 1964).

Result 7. *The reorder point of a swapping station is given by*

$$r \geq T^T\mu - 1 + \Phi^{-1}(1 - \epsilon^S)\sqrt{T^T}\sigma. \tag{15}$$

Result 7 allows us to express r independently from other decision variables, and conveniently incorporate (15) into the infrastructure design optimization problem in the next section. Nevertheless, by ignoring T^D we underestimate r. Therefore, in the second step, we solve that optimization problem, and then adopt the conventional "METRIC" approach (Sherbrooke 2006) to approximate the random retard time T^D by its average $\mathbb{E}[T^D]$. To compute $\mathbb{E}[T^D]$, we can think of the charging station as a queueing system: The orders from the swapping station are "customer arrivals," the released batches of charged batteries are "service completions," and thus the backorders are "waiting customers" (Axsäter 2015). Hence, by Little's law, $\mathbb{E}[T^D]$ equals the ratio of the expected backorder quantity $(\sum_{i\in\mathcal{I}_h} D_i(t) - R)^+$ to the battery demand rate seen by the charging station (Svoronos and Zipkin 1988). This ratio can be numerically evaluated given a solution (r_i, Q_i, R) to the optimization problem. We can then adjust the reorder point r_i for each swapping station i to

$$r_i = \left\lceil (T_i^T + \mathbb{E}[T^D])\mu - 1 + \Phi^{-1}(1 - \epsilon^S)\sqrt{T_i^T + \mathbb{E}[T^D]}\sigma_i \right\rceil \tag{16}$$

(where $\lceil x \rceil$ denotes the smallest integer no smaller than x), and repeat this step if necessary until r_i converges. In fact, our numerical tests suggest that $\mathbb{E}[T^D]$ is so small relative to T_i^T that r_i adjusts insignificantly and never changes planning decisions or managerial insights. For example, Figure 6 shows that $\frac{\mathbb{E}[T^D]}{T_i^T}$ is only 2.5% when $\epsilon^C = 10\%$, and is 8.2% even when ϵ^C is as large as 25%. We thus proceed without considering T^D.

4.3.1. Total Battery Stock Level.

Combining Results 3 and 7 yields the following result.

Result 8. *The total number of batteries in one service network h needs to satisfy*

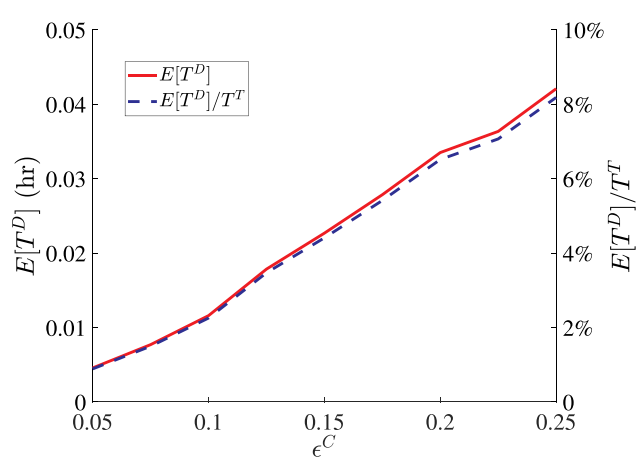
$$R + \sum_{i\in\mathcal{I}_h}(r_i + Q_i) \geq \sum_{i\in\mathcal{I}_h}\left((T^C + 2T_i^T)\mu_i - 1 + Q_i\right) + \Phi^{-1}(1 - \epsilon^C)\sqrt{\sum_{i\in\mathcal{I}_h}\phi_i(Q_i)} + \sum_{i\in\mathcal{I}_h}\Phi^{-1}(1 - \epsilon^S)\sqrt{T_i^T}\sigma_i. \tag{17}$$

It is instructive to compare the minimum total stock level in Result 8 to a base-stock level in the traditional inventory control setting. In the first term of the RHS of (17), $T^C + 2T_i^T$ is akin to a one-period lag in warehouse replenishments to satisfy customer demands at swapping station i, and $(T^C + 2T_i^T)\mu_i - 1$ is the expected demand during one period. However, those battery replenishments have to be from customers. The service network thus also needs to stock Q_i extra batteries to satisfy the initial reorder request from swapping station i, when no batteries returned from customers have yet been available for a second swap. Moreover, the second and the third terms of the RHS of (17) together form the safety stock to cushion against demand uncertainties.

4.4. Comparison with On-Site Charging

We also model the status quo scenario as a benchmark where battery charging is on-site at local swapping stations. In this scenario, we assume that depleted batteries start being charged on-site immediately after being swapped off vehicles. Suppose that the service provider imposes the stockout probability to be no greater than ϵ^B. Then the number of spare batteries r^B should satisfy

Figure 6. (Color online) Mean Order Retard Time $\mathbb{E}[T^D]$ Relative to Transit Time T^T



$\text{Prob}(N^S(T^C) \geq r^B + 1) \leq \epsilon^B$. Approximating $N^S(T^C)$ by a normal random variable matching the mean and the variance, we obtain that the stock level for a swapping station with on-site charging needs to satisfy $r^B \geq T^C\mu - 1 + \Phi^{-1}(1-\epsilon^B)\sqrt{T^C}\sigma$. Hence, the total number of batteries for the swapping stations in subset $\mathcal{I}_h$ would need to satisfy

$$\sum_{i\in\mathcal{I}_h} r_i^B \geq \sum_{i\in\mathcal{I}_h}(T^C\mu_i - 1 + \Phi^{-1}(1-\epsilon^B)\sqrt{T^C}\sigma_i). \tag{18}$$

Batteries are both financially and environmentally costly. Comparing (17) and (18) reveals the following tradeoffs in terms of battery asset heaviness for centralizing versus decentralizing charging:

- *Pooling effect*: Similar to the classical inventory pooling effect, centrally charging batteries leads to a lower level of safety stock $\Phi^{-1}(1-\epsilon^C)\sqrt{\sum_{i\in\mathcal{I}_h}\phi_i(Q_i)}$ than separately building safety stock for each swapping station (i.e., $\sum_{i\in\mathcal{I}_h}\Phi^{-1}(1-\epsilon^C)\sqrt{\phi_i(Q_i)}$). Moreover, as the order quantity Q increases, recall from (5) that the variance of battery deficit, $\phi(Q)$, scales in Q from quadratically to linearly, which also contributes to the pooling effect.
- *Order batching effect:* On the other hand, the main disadvantage of centralized charging is that batching battery orders in groups of $Q > 1$ entails extra batteries. Those extra batteries have to be used for satisfying initial swap demands and for building safety stock (because the variance of battery deficit increases in Q).
- *Lead time effect*: Another disadvantage of centralized charging is the lead time for transporting batteries. As the minimum stock level in (17) shows, the lead time T^T increases the ordering cycle time, and thus elevates both the expected demand per cycle and the safety stock.

We will numerically show in the case study that the favorable pooling effect is dominated by the unfavorable effects of order batching and lead time. In other words, *centralized charging is battery asset-heavier*, which is contrary to the conventional wisdom of pooling inventory. This is because, compared with conventional retail stores, swapping stations with on-site charging are standalone and thus free from order batching or transportation lead time. Hence, the benefit of pooling alone is not enough for justifying the adoption of the "swap-locally, charge-centrally" network. Instead, we will demonstrate in the case study that it is the capacitated urban grid that justifies its adoption.

5. Citywide Infrastructure Deployment

Having characterized the operations of one battery swapping service network, in this section, we propose a joint location and repairable-inventory model along with solution algorithms to study the citywide deployment of the battery swapping service networks.

5.1. Joint Location and Repairable-Inventory Formulation

We first propose a joint location and repairable-inventory model. Given the set of swapping stations $\mathcal{I}$ (which achieves a prespecified level of proximity to customers) and the set of feasible charging locations $\mathcal{H}$, the decision variables include whether to select location $h \in \mathcal{H}$ to build a charging station, denoted by $z_h = 1$ or 0, whether to allocate the swapping station $i \in \mathcal{I}$ to be served by a charging station h, denoted by $y_{ih} = 1$ or 0, and the battery reorder quantity from the charging station h to the swapping station i, denoted by $Q_{ih} \geq 0$.

The objective is to minimize the total amortized cost, including the infrastructure depreciation cost and the battery transportation cost. Let c^C and c^B denote the depreciation costs of a charging station and a battery, respectively. The total depreciation cost is thus given by $\sum_{h\in\mathcal{H}} c_h^C z_h + c^B(\sum_{i\in\mathcal{I}}\sum_{h\in\mathcal{H}}(r_{ih}y_{ih} + Q_{ih}) + \sum_{h\in\mathcal{H}} R_h)$. The second term is from Results 7 and 8: The reorder point of swapping station i served by charging station h is $r_{ih} = T_{ih}^T\mu_i - 1 + \Phi^{-1}(1-\epsilon_i^S)\sqrt{T_{ih}^T}\sigma_i$, and the battery stock level of service network h can be rewritten as

$$R_h = \sum_{i\in\mathcal{I}}(T_{ih}^T + T^C)\mu_i y_{ih} + \Phi^{-1}(1-\epsilon_h^C)\sqrt{\sum_{i\in\mathcal{I}} y_{ih}\phi_{ih}(Q_{ih})}.$$

Moreover, the mean cost of shipping batteries between charging and swapping stations is $2c_{ih}^T\sum_{t\in\mathcal{T}}\kappa_t\mu_{it}y_{ih}/Q_{ih}$, where c_{ih}^T is the one-way transportation cost, and κ_t is the proportion of time in period $t \in \mathcal{T}$. Here we estimate the long-run average rate of battery shipments to and from a charging station as twice the average swap demand rate $\sum_{t\in\mathcal{T}}\kappa_t\mu_{it}$.

Collectively, the total cost as a function of the decision variables is given by

$$\begin{aligned} C(\boldsymbol{z},\boldsymbol{y},\boldsymbol{Q}) &= \sum_{h\in\mathcal{H}} c_h^C z_h + c^B\left(\sum_{i\in\mathcal{I}}\sum_{h\in\mathcal{H}}(r_{ih}y_{ih} + Q_{ih}) + \sum_{h\in\mathcal{H}} R_h\right) \\ &\quad + \sum_{i\in\mathcal{I}}\sum_{h\in\mathcal{H}}\frac{2c_{ih}^T\sum_{t\in\mathcal{T}}\kappa_t\mu_{it}y_{ih}}{Q_{ih}} \\ &= \sum_{h\in\mathcal{H}} c_h^C z_h + \sum_{h\in\mathcal{H}}\sum_{i\in\mathcal{I}}\alpha_{ih}y_{ih} \\ &\quad + \sum_{h\in\mathcal{H}}\sum_{i\in\mathcal{I}}\left(c^B Q_{ih} + \frac{\beta_{ih}y_{ih}}{Q_{ih}}\right) \\ &\quad + \sum_{h\in\mathcal{H}}\gamma_h\sqrt{\sum_{i\in\mathcal{I}} y_{ih}\phi_{ih}(Q_{ih})}, \end{aligned} \tag{19}$$

where $\alpha_{ih} \triangleq c^B((2T_{ih}^T + T^C)\mu_i - 1 + \Phi^{-1}(1-\epsilon_i^S)\sqrt{T_{ih}^T}\sigma_i)$, $\beta_{ih} \triangleq 2c_{ih}^T\sum_{t\in\mathcal{T}}\kappa_t\mu_{it}$, and $\gamma_h \triangleq c^B\Phi^{-1}(1-\epsilon_h^C)$ are parameters for notation brevity. Subsequently, the problem of deploying battery swapping service networks can be formulated as the following program, which we term as a

joint location and repairable-inventory model:

$$\text{(P)} \quad \min_{z,y,Q} \; C(z,y,Q) \tag{20a}$$

$$\text{s.t.} \quad y_{ih} \le z_h, \qquad \forall i \in \mathcal{I},\ \forall h \in \mathcal{H}, \tag{20b}$$

$$\sum_{h\in\mathcal{H}} y_{ih} = 1, \qquad \forall i \in \mathcal{I}, \tag{20c}$$

$$y_{ih}, z_h \in \{0,1\}, \qquad \forall i \in \mathcal{I},\ \forall h \in \mathcal{H}, \tag{20d}$$

$$0 \le Q_{ih} \le \overline{Q} y_{ih}, \qquad \forall i \in \mathcal{I},\ \forall h \in \mathcal{H}. \tag{20e}$$

In the previous formulation, Constraints (20b) and (20c) ensure that each swapping station is served by one open charging station. Constraint (20e) bounds from the above the reorder quantity Q_{ih} by the truckload capacity $\overline{Q}$. Although $\phi_{ih}(Q_{ih})$ is a convex function in $Q_{ih} \in [0, +\infty)$ as (5) suggests, $\sqrt{\sum_{i\in\mathcal{I}} y_{ih}\phi_{ih}(Q_{ih})}$ is neither convex nor convex in Q because $\phi_{ih}(Q_{ih})$ is convex quadratic when $0 \le Q_{ih} \le \nu_{ih}$ but linear when $Q_{ih} \ge \nu_{ih}$. Therefore, (P) does not permit a convex mixed integer reformulation and thus cannot be directly solved by existing commercial solvers. In the following section, we design an effective algorithm to solve (P) exactly by exploiting its structural properties. We also provide in Online Appendix F an alternative formulation that incorporates random charging time, to which this algorithm also applies.

5.2. Constraint Generation and Parameter Search Algorithm

This section proposes an efficient algorithm to solve (P) exactly. The main idea is as follows: First, we show that (P) can be reformulated into a constrained submodular minimization problem where the nonconvex nonconcave structure is encapsulated into submodular cost terms. Second, we use the Lovász extension of submodular functions to design an exact constraint generation (CG) algorithm. The master problem of the CG algorithm is a linear mixed 0-1 optimization problem and thus can be solved efficiently by existing solvers. Third, to check the stopping criterion or generate new constraints for the master problem, we need to evaluate these submodular cost terms by solving a series of nonlinear evaluation subproblems. As another algorithmic contribution of this paper, we solve these problems efficiently by designing a subalgorithm based on the idea of parametric search (PS). The remainder of this subsection unfolds this CG&PS algorithm.

5.2.1. Constrained Submodular Minimization Reformulation.

Problem (P) can be equivalently reformulated into the following nonlinear facility location problem:

$$\min_{z,y} \quad f(z,y) = \sum_{h\in\mathcal{H}} c_h^C z_h + \sum_{h\in\mathcal{H}}\sum_{i\in\mathcal{I}} \alpha_{ih} y_{ih} + \sum_{h\in\mathcal{H}} \gamma_h \cdot g_h(\boldsymbol{y_h})$$

$$\text{s.t.} \quad \text{(20b)–(20d)}. \tag{21}$$

In (21), $\boldsymbol{y_h} = (y_{ih} : i \in \mathcal{I}) \in \{0,1\}^{|\mathcal{I}|}$ is the allocation vector for charging station h, and $g_h(\boldsymbol{y_h})$ is a nonlinear function of $\boldsymbol{y_h}$ when each swapping station served by charging station h (i.e., $y_{ih} = 1$) chooses the optimal reorder point. Mathematically,

$$g_h(\boldsymbol{y_h}) = \min_{0\le Q_{ih}\le \overline{Q} y_{ih},\, \forall i\in\mathcal{I}} \left\{ \sqrt{\sum_{i\in\mathcal{I}} y_{ih}\phi_{i,h}(Q_{ih})} + \sum_{i\in\mathcal{I}} \left(a_h Q_{ih} + \frac{b_{ih} y_{ih}}{Q_{ih}} \right) \right\},$$

where $a_h \triangleq \frac{c^B}{\gamma_h} \ge 0$ and $b_{ih} \triangleq \frac{\beta_{ih}}{\gamma_h} > 0$.

We next show that, for any $h \in \mathcal{H}$, $g_h(\cdot)$ is a submodular function over $\{0,1\}^{|\mathcal{I}|}$. Consider the following set function $g : 2^{|\mathcal{I}|} \to R$ such that $g(\emptyset) = 0$. For any $S \subseteq \mathcal{I}$ and $S \ne \emptyset$, define $g(S)$ by the following evaluation subproblem (index h is omitted):

$$(\text{P}_S) \quad g(S) \triangleq \min_{Q} \left\{ \sqrt{\sum_{i\in S} \phi_i(Q_i)} + \sum_{i\in S} \left(aQ_i + \frac{b_i}{Q_i} \right) : 0 \le Q_i \le \overline{Q},\ \forall i \in S, Q_i = 0,\ \forall i \in \mathcal{I} \setminus S \right\}.$$

The following lemma establishes the submodularity of g.

Lemma 8. *The function $g(\cdot) : 2^{|\mathcal{I}|} \to R$ is a submodular function.*

5.2.2. Mixed 0-1 Reformulation.

The submodularity of g allows us to extend g to a continuous convex function based on the Lovász extension and propose a mixed 0-1 reformulation for (P).

To define the Lovász extension, we introduce the following notation. Let Π be the set of permutations of elements in $\mathcal{I}$. For any $\pi \in \Pi$, let S_j^π be the set of the first jth elements in permutation π. Let π_j be the jth element in π, and $[i]$ be the position of element i in π, that is, $\pi_{[i]} = i$. For a set function $g : 2^n \to R$, $\pi \in \Pi$ and $j \in \{1, \ldots, |\mathcal{I}|\}$, define

$$g_j^\pi = g(S_j^\pi) - g(S_{j-1}^\pi), \quad g(S_0^\pi) = 0.$$

Then the Lovász extension of submodular functions leads to the following result.

Lemma 9. *For any $\boldsymbol{y} \in \{0,1\}^{|\mathcal{I}|}$, the submodular function g satisfies:*

(i) $g(\boldsymbol{y}) = \max_{\pi\in\Pi} \sum_{i\in\mathcal{I}} g_{[i]}^\pi y_i$.

(ii) $g(\boldsymbol{y}) = \sum_{i\in\mathcal{I}} g_{[i]\overline{\pi}} y_i$, *where* $\overline{\pi} \in \Pi$ *satisfies* $y_{\overline{\pi}_1} \ge y_{\overline{\pi}_2} \ge \cdots \ge y_{\overline{\pi}_{|\mathcal{I}|}}$.

Lemma 9 shows that $g(\boldsymbol{y})$ can be expressed as a linear combination of $\{y_i : i \in \mathcal{I}\}$, in which the coefficients $g_{[i]}^\pi$ are given by a permutation π that maximizes the function value. Moreover, for any $\boldsymbol{y} \in \{0,1\}^{|\mathcal{I}|}$, we can easily find a $\overline{\pi}$ by sorting $\{y_i : i \in \mathcal{I}\}$ such that $g(\boldsymbol{y}) = \sum_{i\in\mathcal{I}} g_{[i]\overline{\pi}} y_i$.

From Lemma 9, (P) can be further reformulated as a mixed 0-1 linear program:

$$\text{(MIP)} \quad \min_{z,y,w} \quad \sum_{h\in\mathcal{H}} c_h^C z_h + \sum_{h\in\mathcal{H}}\sum_{i\in\mathcal{I}} \alpha_{ih} y_{ih} + \sum_{h\in\mathcal{H}} \gamma_h w_h \tag{22a}$$

$$\text{s.t.} \quad (20b)-(20d),$$

$$w_h \geq \sum_{i\in\mathcal{I}} g^{\pi}_{[i],h} y_{i,h}, \quad \forall h\in\mathcal{H},\ \forall \pi\in\Pi, \tag{22b}$$

where w_h is an auxiliary variable; $g^{\pi}_{i,h} = g_h(S^{\pi}_i) - g_h(S^{\pi}_{i-1})$ and $g_h(S^{\pi}_0) = 0$.

(MIP) eliminates the nonlinear terms in (P), but incurs two major difficulties. First, (MIP) has an exponential number ($|\mathcal{I}|$ factorial) of Constraints (22b). Second, specifying the RHS of each Constraint (22b) requires solving $|\mathcal{I}|$ evaluation subproblems (P_S) to obtain coefficients $g^{\pi}_{[i],h}$, which is nontrivial given the nonconvexity of (P_S). In what follows, we overcome the first difficulty by proposing a constraint generation algorithm and then address the second difficulty by developing an exact customized parameter search algorithm.

5.2.3. Constraint Generation Algorithm. The constraint generation algorithm solves (MIP) by iteratively solving the following *master problem* with a subset of constraints:

$$(\text{MP}_k) \quad \min_{z,y,w} \quad \sum_{h\in\mathcal{H}} c_h^C z_h + \sum_{h\in\mathcal{H}}\sum_{i\in\mathcal{I}} \alpha_{ih} y_{ih} + \sum_{h\in\mathcal{H}} \gamma_h w_h$$

$$\text{s.t.} \quad (20b)-(20d),$$

$$w_h \geq \sum_{i\in\mathcal{I}} g^{\pi}_{[i],h} y_{i,h}, \quad \forall h\in\mathcal{H},\ \forall \pi\in\Pi^k_h,$$

where $\Pi^k_h \subseteq \Pi$ is only a subset of permutations for charging station h. Hence, in general, (MP_k) is significantly easier to solve than (MIP).

An overview of the constraint generation algorithm is as follows. Let $(\boldsymbol{z^k}, \boldsymbol{y^k}, \boldsymbol{w^k})$ be the optimal solution to (MP_k). Let f^k and f^* be the optimal objective values of (MP_k) and (MIP), respectively. Because (MP_k) is a relaxed problem of (MIP), it follows that $f^k \leq f^*$. Furthermore, we check if $w^k_h \geq g_h(\boldsymbol{y^k_h})$ holds for all $h\in\mathcal{H}$ by solving the associated evaluation subproblems. If true, then Lemma 9(i) ensures that $w^k_h \geq \max_{\pi\in\Pi}\sum_{i\in\mathcal{I}} g^{\pi}_{[i]} y_i$ also holds for all $h\in\mathcal{H}$. In that case, $(\boldsymbol{z^k}, \boldsymbol{y^k}, \boldsymbol{w^k})$ is also feasible to (MIP) and thus $f^k \geq f^*$, which implies that $(\boldsymbol{z^k}, \boldsymbol{y^k}, \boldsymbol{w^k})$ is optimal for (MIP). Otherwise, there must be some $\hat{h}\in\mathcal{H}$ such that $w^k_{\hat{h}} < g_{\hat{h}}(\boldsymbol{y^k_{\hat{h}}})$. For each $\hat{h}$, we generate a permutation $\hat{\pi}\in\Pi$ by sorting $\{y^k_{i,\hat{h}} : i\in\mathcal{I}\}$ in the nonincreasing order so that $y^k_{\hat{\pi}_1,\hat{h}} \geq y^k_{\hat{\pi}_2,\hat{h}} \cdots \geq y^k_{\hat{\pi}_{|\mathcal{I}|},\hat{h}}$. We update $\Pi^{k+1}_{\hat{h}} = \Pi^k_{\hat{h}} \cup \{\hat{\pi}\}$ and add the associated constraint into (MP_k). From Lemma 9(ii), $g_{\hat{h}}(\boldsymbol{y^k_{\hat{h}}}) = \sum_{i\in\mathcal{I}} g^{\hat{\pi}}_{[i],\hat{h}} y^k_{i,\hat{h}}$, and thus $(\boldsymbol{z^k}, \boldsymbol{y^k}, \boldsymbol{w^k})$ will be cut off in the next iteration. Hence, we obtain an updated f^k as a tightened lower bound of f^*. Meanwhile, in each iteration, we update an upper bound of f^*, denoted by $\bar{f}$. This is achievable by observing that $(\boldsymbol{z^k}, \boldsymbol{y^k})$ is feasible to Reformulation (21), and thus the objective value $f(\boldsymbol{z^k}, \boldsymbol{y^k})$ is an upper bound of (P). The entire algorithm is summarized in Algorithm 1, which returns a feasible solution to (P) with an optimality gap less than a prespecified relative tolerance level $\varepsilon_{\text{CG}} > 0$, that is, $\frac{\bar{f}-f^*}{f^*} \leq \varepsilon_{\text{CG}}$.

Algorithm 1 (Constraint Generation Algorithm for (P))

Input: $\varepsilon_{\text{CG}} > 0$

Output: $(\bar{\boldsymbol{z}}, \bar{\boldsymbol{y}})$ and $\bar{f}$.

1: Initialize $k = 1$, $\bar{f} = +\infty$, and $\Pi^1_h = \{(1, 2, \ldots, |\mathcal{I}|)\}$ for $h\in\mathcal{H}$.
2: Solve (MP_k) to obtain $(\boldsymbol{z^k}, \boldsymbol{y^k}, \boldsymbol{w^k})$ and f^k.
3: If $\bar{f} \geq f(\boldsymbol{z^k}, \boldsymbol{y^k})$, then $\bar{f} = f(\boldsymbol{z^k}, \boldsymbol{y^k})$ and $(\bar{\boldsymbol{z}}, \bar{\boldsymbol{y}}) = (\boldsymbol{z^k}, \boldsymbol{y^k})$.
4: If $\frac{\bar{f}-f^k}{f^k} \leq \varepsilon_{\text{CG}}$, then return $(\bar{\boldsymbol{z}}, \bar{\boldsymbol{y}})$ and $\bar{f}$; else, goto Step 5.
5: For any $\hat{h}\in\mathcal{H}$ such that $w^k_{\hat{h}} < g_{\hat{h}}(\boldsymbol{y^k_{\hat{h}}})$, generate $\hat{\pi}\in\Pi$ by sorting $\{y^k_{i,\hat{h}} : i\in\mathcal{I}\}$ in the nonincreasing order. Compute $g_{\hat{h}\hat{\pi}}$ and update $\Pi^{k+1}_{\hat{h}} = \Pi^k_{\hat{h}} \cup \{\hat{\pi}\}$. Goto step 2.

5.2.4. Parameter Search Algorithm for Evaluation Subproblem (P_S). Algorithm 1 requires solving the evaluation subproblem (P_S) to update the upper bound $\bar{f}$ in Step 3 and to evaluate and generate Constraints (22b) in Step 5. As previously mentioned, (P_S) is a multidimensional nonconvex nonconcave minimization problem. In general, given any $S \neq \emptyset$, (P_S) cannot be directly solved by solvers. To tackle this difficulty, we next propose an exact customized parameter search algorithm.

We describe this algorithm with the following notation. In (P_S), let $u(\boldsymbol{Q}) \triangleq \sum_{i\in S}\phi_i(Q_i)$ and $v(\boldsymbol{Q}) \triangleq \sum_{i\in S}\left(aQ_i + \frac{b_i}{Q_i}\right)$. Define $H \triangleq \{(u,v) | u = u(\boldsymbol{Q}), v = v(\boldsymbol{Q}), 0 \leq Q_i \leq \overline{Q},\ \forall i\in S\}$. Let $\text{conv}(H)$ be the convex hull of H. Hence, (P_S) can be reformulated as

$$\begin{aligned} g(S) &= \min\left\{\sqrt{u(\boldsymbol{Q})} + v(\boldsymbol{Q}) : 0 \leq Q_i \leq \overline{Q},\ \forall i\in S\right\} \\ &= \min\{\sqrt{u} + v : (u,v)\in H\} \\ &= \min\{\sqrt{u} + v : (u,v)\in \text{conv}(H)\}, \end{aligned}$$

where the last equality is because the function $\sqrt{u} + v$ is concave in (u, v) and attains its minimum at an extreme point of $\text{conv}(H)$. Therefore, there exists an optimal solution $\boldsymbol{Q}^*$ of (P_S), such that $(u(\boldsymbol{Q}^*), v(\boldsymbol{Q}^*))$ is an extreme point of $\text{conv}(H)$. Earlier works such as Shen et al. (2003) and Shu et al. (2005) have designed elegant polynomial-time algorithms to solve *unconstrained discrete* concave minimization problems, and Zhang et al. (2018) propose an efficient algorithm to solve a *constrained discrete* quasi-convex maximization problem based on the idea of parametric search. However, the problems they consider include either a polynomial number or a finite number of extreme points, whereas

here conv(H) has an infinite number of extreme points. To design an efficient customized algorithm, we further exploit the separable structure of the parameterized subproblem and the monotonicity of the critical parameter over a series of evaluation subproblems.

Specifically, note that the extreme points of conv(H) can be obtained by minimizing a linear function over conv(H). Therefore, we consider the following parameterized subproblem, which minimizes a parameterized linear function ($\theta \geq 0$) over conv(H):

$$(\mathrm{P}_\theta) \quad \min\{\theta u + v : (u,v) \in \mathrm{conv}(H)\}$$
$$= \min\{\theta u(\boldsymbol{Q}) + v(\boldsymbol{Q}) : 0 \leq Q_i \leq \overline{Q},\ \forall i \in S\}.$$

Let $\boldsymbol{Q}_\theta$ be the optimal solution to (P_θ). Define $u_\theta = u(\boldsymbol{Q}_\theta)$ and $v_\theta = v(\boldsymbol{Q}_\theta)$.

The proposed algorithm is based on the following result.

Proposition 2. *The following properties hold:*

(i) *Let $\boldsymbol{Q}^*$ be the optimal solution to* (P_S) *and* $\theta^* = \frac{1}{2\sqrt{u(\boldsymbol{Q}^*)}}$. *Then any optimal solution $\boldsymbol{Q}_{\theta^*}$ to* (P_{θ^*}) *is also optimal for* (P_S).

(ii) *Given any $0 < \theta_1 < \theta_2$, let $(\hat{u}, \hat{v})$ be the intersection point of lines $l_j : v = \theta_j(u_{\theta_j} - u) + v_{\theta_j}$ ($j = 1, 2$). Let $g_{\theta_1,\theta_2} = \min\{\sqrt{\hat{u}} + \hat{v}, \sqrt{u_{\theta_1}} + v_{\theta_1}, \sqrt{u_{\theta_2}} + v_{\theta_2}\}$. Then, any $\theta \in [\theta_1, \theta_2]$ satisfies*

$$\sqrt{u_\theta} + v_\theta \geq g_{\theta_1,\theta_2}.$$

Proposition 2(i) shows the existence of the critical parameter value θ^*. To solve (P_S), it suffices to search for θ^* and solve (P_{θ^*}), which we will show is easier to solve due to its separable structure in Lemma 10. Proposition 2(ii) provides g_{θ_1,θ_2} as a lower bound of $\sqrt{u_\theta} + v_\theta$ over any interval $[\theta_1, \theta_2]$ and thus gives a necessary condition for θ^* to be contained in an interval $[\theta_1, \theta_2]$, that is, $\sqrt{u_{\theta^*}} + v_{\theta^*}$ must be no smaller than g_{θ_1,θ_2}; otherwise, we know that $[\theta_1, \theta_2]$ does not contain θ^*.

Proposition 2 paves the way for a branch-and-bound based interval search to find the critical parameter θ^*. Specifically, the search initializes with the active interval set $\mathcal{M}$ being a singleton $\{[\underline{\theta}, \overline{\theta}]\}$ (we will describe how to set $\underline{\theta}$ and $\overline{\theta}$ in Lemma 11). As long as $\mathcal{M} \neq \emptyset$, we select an interval from $\mathcal{M}$, label it as $[\theta_1, \theta_2]$, and check if the incumbent best upper bound of the optimal objective value of (P_S) is smaller than g_{θ_1,θ_2}. If true, we can prune $[\theta_1, \theta_2]$ from $\mathcal{M}$ due to Proposition 2(ii). Otherwise, we branch $[\theta_1, \theta_2]$ into two intervals $[\theta_1, \hat{\theta}]$ and $[\hat{\theta}, \theta_2]$ and update $\mathcal{M} = \mathcal{M} \cup \{[\theta_1, \hat{\theta}], [\hat{\theta}, \theta_2]\}$. The partition point $\hat{\theta}$ can be set as $\frac{v_{\theta_2} - v_{\theta_1}}{u_{\theta_1} - u_{\theta_2}}$, which is the root of equation $\theta u_{\theta_1} + v_{\theta_1} = \theta u_{\theta_2} + v_{\theta_2}$. Because (u_θ, v_θ) minimizes $\theta u_\theta + v_\theta$ for any θ, it follows that $\theta_1 u_{\theta_1} + v_{\theta_1} \leq \theta_1 u_{\theta_2} + v_{\theta_2}$ and $\theta_2 u_{\theta_1} + v_{\theta_1} \geq \theta_2 u_{\theta_2} + v_{\theta_2}$, which implies that $\hat{\theta} \in [\theta_1, \theta_2]$. This branch-and-bound process repeats until $\mathcal{M}$ becomes an empty set. Then the algorithm returns θ^* and the optimal solution $\boldsymbol{Q}^* = \boldsymbol{Q}_{\theta^*}$ (due to Proposition 2(i)).

Having outlined the main procedure of the parameter search algorithm, we next describe three subroutines. These customized subroutines exploit the problem structure and are critical to enhancing computational efficiency.

(1) *How to solve* (P_θ). The algorithm needs to frequently solve (P_θ) for $\boldsymbol{Q}_\theta$ to obtain u_θ, v_θ, $\sqrt{u_\theta} + v_\theta$ and g_{θ_1,θ_2}. We next show that (P_θ) permits a closed-form solution $\boldsymbol{Q}_\theta$. For any $h \in \mathcal{H}$ and $\theta \geq 0$, (P_θ) can be decomposed into $|S|$ univariate convex optimization problems:

$$(\mathrm{P}_{\theta,i}) \quad \min\left\{\theta\phi_i(Q_i) + aQ_i + \frac{b_i}{Q_i} : 0 \leq Q_i \leq \overline{Q}\right\}, \quad \forall i \in S.$$

Each ($\mathrm{P}_{\theta,i}$) can be further segmented into two problems (recall $\phi_i(Q_i)$ is given by (5)):

$$(\mathrm{P}^1_{\theta,i}) \quad \min\left\{\frac{\theta}{6}Q_i^2 + aQ_i + \frac{b_i}{Q_i} : 0 \leq Q_i \leq \nu_i\right\} + \xi_i\theta,$$

$$(\mathrm{P}^2_{\theta,i}) \quad \min\left\{(a + \iota_i\theta)Q_i + \frac{b_i}{Q_i} : \nu_i \leq Q_i \leq \overline{Q}\right\} - \iota_i^2\theta,$$

where $\xi_i \triangleq (T_i^T + T^C)\sigma_i^2 - 1/6$, $\iota_i \triangleq (T_i^T + T^C)\mu_i$ and $\nu_i = 3\iota_i - \sqrt{3\iota_i^2 - 6\xi_i} \geq 3\iota_i$. Without loss generality, suppose $\nu_i \leq \overline{Q}$; otherwise, only ($\mathrm{P}^1_{\theta,i}$) is relevant.

Let $Q_i(\theta)$, $Q_{i,1}(\theta)$ and $Q_{i,2}(\theta)$ be the optimal solution to ($\mathrm{P}_{\theta,i}$), ($\mathrm{P}^1_{\theta,i}$) and ($\mathrm{P}^2_{\theta,i}$), respectively. The following lemma characterizes $Q_{i,1}(\theta)$, $Q_{i,2}(\theta)$ and $Q_i(\theta)$.

Lemma 10. *For all $\theta \geq 0$, $i \in S$,*

(i) $Q_{i,1}(\theta) = \min\{\tilde{Q}_{\theta,i}, \nu_i\}$, *where $\tilde{Q}_{\theta,i}$ is the unique nonnegative real root of the cubic equation* $\frac{\theta Q^3}{3} + aQ^2 - b_i = 0$.

(ii) $Q_{i,2}(\theta) = \min\{\max\{\sqrt{b_i/(a + \iota_i\theta)}, \nu_i\}, \overline{Q}\}$.

(iii) $Q_i(\theta) = \tilde{Q}_{\theta,i}$, *if* $\sqrt{b_i/a} \leq \nu_i$ *or* $\theta > \overline{\theta}_{i,1}$, *where* $\overline{\theta}_{i,1} = \frac{3}{\nu_i}\left(\frac{b_i}{\nu_i^2} - a\right)$; *otherwise,* $Q_i(\theta) = Q_{i,2}(\theta)$.

(iv) $Q_{i,1}(\theta)$, $Q_{i,2}(\theta)$ *and* $Q_i(\theta)$ *are nonincreasing in* θ.

Lemma 10, (i)–(iii), shows that (P_θ) can be solved analytically (the root $\tilde{Q}_i$ can be directly computed by the Cardano's formula). Moreover, Lemma 10(iv) establishes the monotonicity of the optimal solutions in θ, which is used in algorithm initialization and speedup. A graph illustrating how $Q_i(\theta)$ varies in θ is available in Online Appendix D.4 along with the proof.

(2) *Initialization*. The parameter search algorithm starts with the initial bounds $\underline{\theta}$ and $\overline{\theta}$ for θ^*. The following lemma provides these bounds.

Lemma 11. *For any $i \in S$, let $\tilde{p}_i$ be the unique nonnegative real root of equation* $\frac{1}{36}p^3 = (\xi_i + 1/6p)(b_i - ap)^2$. *Then*

$$Q_i^L \triangleq \min\left\{\sqrt{\tilde{p}_i}, \nu_i\} \leq Q_i^* \leq \min\{\sqrt{b_i/a}, \overline{Q}\right\} \triangleq Q_i^U,$$

$$\underline{\theta} \triangleq \frac{1}{2\sqrt{u(\boldsymbol{Q}^U)}} \leq \theta^* \leq \frac{1}{2\sqrt{u(\boldsymbol{Q}^L)}} \triangleq \overline{\theta}.$$

In addition to providing the initial bounds $\underline{\theta}$ and $\overline{\theta}$ for θ^*, Lemma 11 can also be used to tighten problem (P_θ) by shrinking the feasible region for each Q_i from $[0, \overline{Q}]$ to $[Q_i^L, Q_i^U]$.

(3) *Speedup.* In the constraint generation algorithm, generating each constraint requires solving a series of evaluation problems for a given permutation $\pi \in \Pi$. We can exploit the structural connection between these evaluation problems to speed up the algorithm.

To simplify the notation, consider $\pi = (1, 2, \ldots, |\mathcal{I}|)$. Let $S_k = \{1, 2, \ldots, k\}$ for $k \in \mathcal{I}$. Let $\boldsymbol{Q}^k$ and θ^k be the optimal solution and the critical parameter for (P_{S_k}). The following proposition shows that $\boldsymbol{Q}^k$ is nondecreasing in k, whereas θ^k is nonincreasing in k.

Proposition 3. *For any* $k = 1, \ldots, |\mathcal{I}| - 1$, (i) $Q_i^k \le Q_i^{k+1}$, $\forall i \in \mathcal{I}$, *and* (ii) $\theta^k \ge \theta^{k+1}$.

Proposition 3 can be used to accelerate the parameter search algorithm for multiple evaluation subproblems associated with a given permutation. Suppose we have solved (P_S) and (P_T). Now consider (P_R) such that $S \subseteq R \subseteq T$. The lower and upper bounds for θ^* of (P_R) can thus be tightened as $\frac{1}{2\sqrt{u(Q^T)}} \le \theta^* \le \frac{1}{2\sqrt{u(Q^S)}}$ due to Proposition 2(i) and 3.

The pseudocode of this parameter search algorithm is summarized in Algorithm 2, which returns a solution $\boldsymbol{Q}^*$ to (P_S) within a prespecified relative gap tolerance $\varepsilon_{\mathrm{PS}} > 0$. When solving multiple evaluation subproblems associated with a given permutation, the optimal solution $\boldsymbol{Q}^*$ to the previous subproblem is used to obtain tightened initial interval $[\underline{\theta}, \overline{\theta}]$ for the subsequent subproblem. Although the worst-case computational complexity of this algorithm may be nonpolynomial (as (P_S) is a nonconvex minimization problem), our extensive experiments (reported in Section 6.1) show that the customized parameter search algorithm is surprisingly efficient. For all test instances, no more than three intervals are explored even if $\varepsilon_{\mathrm{PS}}$ is as small as 10^{-7}.

Algorithm 2 (Parameter Search Algorithm for (P_S))

Input: $\varepsilon_{\mathrm{PS}} > 0$

Output: $\boldsymbol{Q}^*$ and g^*

1: Initialize $\mathcal{M} = \{[\underline{\theta}, \overline{\theta}]\}$. Compute $(u_{\underline{\theta}}, v_{\underline{\theta}})$ and $(u_{\overline{\theta}}, v_{\overline{\theta}})$.
If $\sqrt{u_{\underline{\theta}}} + v_{\underline{\theta}} \le \sqrt{u_{\overline{\theta}}} + v_{\overline{\theta}}$, set $\boldsymbol{Q}^* \leftarrow \boldsymbol{Q}(\underline{\boldsymbol{\theta}})$; else, $\boldsymbol{Q}^* \leftarrow \boldsymbol{Q}(\overline{\boldsymbol{\theta}})$. Set $g^* \leftarrow \sqrt{u(\boldsymbol{Q}^*)} + v(\boldsymbol{Q}^*)$.

2: If $\mathcal{M} = \emptyset$, return current $(\boldsymbol{Q}^*, g^*)$. Else, select an interval $[\theta_1, \theta_2]$ from $\mathcal{M}$. Delete this interval from $\mathcal{M}$.

3: Compute the lower bound g_{θ_1,θ_2}. If $\frac{g^* - g_{\theta_1,\theta_2}}{g_{\theta_1,\theta_2}} < \varepsilon_{\mathrm{PS}}$, goto Step 2; else, goto Step 4.

4: Set $\hat{\theta} \leftarrow \frac{v_{\theta_2} - v_{\theta_1}}{u_{\theta_1} - u_{\theta_2}}$. Augment $\mathcal{M} \leftarrow \mathcal{M} \cup \{[\theta_1, \hat{\theta}], [\hat{\theta}, \theta_2]\}$. Compute $(u_{\hat{\theta}}, v_{\hat{\theta}})$. If $g^* \ge \sqrt{u_{\hat{\theta}}} + v_{\hat{\theta}}$, set $g^* \leftarrow \sqrt{u_{\hat{\theta}}} + v_{\hat{\theta}}$ and $\boldsymbol{Q}^* \leftarrow \boldsymbol{Q}(\hat{\theta})$. Goto Step 2.

Remark. Our proposed algorithmic framework (i.e., constraint generation and parameter search) may potentially have broader applications. Recent advances in literature have extended the utilization of submodularity from the restricted 0-1 minimization settings to a more general set of optimization problems that involve, for example, conic quadratic mixed 0-1 minimization, to improve computational efficiency (Atamtürk and Gómez 2020). However, whereas those problems are solvable per se by commercial solvers, there remain many other problems that intrinsically do not permit convex reformulations despite having submodularity (and are less studied in the literature). Beyond battery swapping, those problems may extensively arise in stochastic inventory-location models and queueing-location models where additional decisions (such as order quantities of retailers and local service capacity) may give rise to nonconvex and piecewise minimization subproblems. We envision that our algorithmic framework can be utilized to tackle those problems.

6. Case Study and Managerial Implications

Given the planning model and the proposed solution algorithm in Section 5, the service provider can evaluate whether and how to scale up battery swapping service capacity citywide. This section tests the computational performance of the solution algorithm and then presents managerial insights. Because of the page limit, we report detailed model calibration in Online Appendix J.

6.1. Algorithm Efficacy

We first demonstrate the computational effectiveness of the constraint generation and parameter search (CG&PS) algorithm proposed in Section 5.2 for solving the citywide battery swapping service network deployment problem (P).

6.1.1. Setup. We generate additional parameters for (P) based on the gas station data in Beijing. In particular, the locations of swapping stations and candidate charging stations are randomly sampled from the locations of the 133 gasoline refueling stations in Beijing with $|\mathcal{I}| \in \{30, 60, 90\}$ and $|\mathcal{H}| \in \{1/3, 1/2, 2/3\}|\mathcal{I}|$. For any given problem size $(|\mathcal{I}|, |\mathcal{H}|)$, we randomly generate 50 problem instances where the charging station cost c^C and the mean and standard deviation of hourly swap demands (μ and σ) randomly vary from 75% to 125% of their baseline values.

6.1.2. Benchmark 1: Gurobi. We benchmark the proposed algorithm against the state-of-the-art commercial solver Gurobi. However, recall that (P) and the evaluation subproblem (P_S) are nonconvex, piecewise, and thus cannot be directly solved by Gurobi. Therefore, we additionally generate *convexified* problem instances such that they reduce to convex optimization. Specifically, we modify each instance by setting the truckload capacity $\overline{Q} = \nu_i$ for all $i \in \mathcal{I}$ so that the piecewise function

$\phi(Q)$ becomes a convex quadratic function over $[0, \overline{Q}]$. For these convexified problem instances, (P) allows a mixed integer conic programming reformulation and (P_S) reduces to a continuous convex optimization problem. See Online Appendix G for the detailed reformulation.

6.1.3. Benchmark 2: Branch-and-Price Algorithm. We also construct a branch-and-price (BP) algorithm as the second benchmark. The main idea is as follows: We reformulate (P) as a set-partitioning problem and solve the corresponding linear program relaxation using the column generation method. Given any set of dual prices from this relaxed problem, we need to solve a *pricing subproblem* to quickly identify a column with negative reduced cost or to verify that no such column exists. Existing papers such as Teo and Shu (2004) and Ni et al. (2021) have used this BP procedure to solve integrated facility location problems. However, existing methods cannot be directly applied to solve the pricing subproblem in our setting. We thus invoke our proposed CG&PS algorithm. A more detailed description is available in Online Appendix H.

6.1.4. Benchmark 3: Heuristic Algorithm. For the purposes of both providing another benchmark and providing practitioners with an easy-to-implement solution approach, we additionally develop a customized local-search and iterative improvement algorithm to solve (P). This heuristic algorithm also uses the proposed PS algorithm to solve the evaluation subproblems. More details of this heuristic algorithm are available in Online Appendix I.

In total, we solve 1,800 $(= 3 \times 3 \times 50 \times 4)$ instances. All computational studies are carried out in MATLAB running on a Dell desktop with an Intel Core i7 CPU (2.4 GHz) and 32 GB RAM. Throughout the experiments, we set $\varepsilon_{PS} = 10^{-7}$ and $\varepsilon_{CG} = 10^{-4}$. For the convexified instances of (P_S) and (P), the relative gap tolerances of the Gurobi 9.1.0 solver are to set at 10^{-7} and 10^{-4}, respectively. In the CG procedure, Gurobi is used to solve the master problem (MP_k) with the relative gap tolerance 10^{-6}. The running time limit for all instances is set at 1,800 seconds.

6.1.5. Performance of Parameter Search. We first examine the performance of the PS algorithm for solving (P_S). For all test instances, the PS algorithm only needs to explore a very few number of intervals and takes no more than 0.1 milliseconds on average to solve an evaluation subproblem. For the convexified problems, our algorithm is capable of reducing the computational time of the Gurobi solver by at least three orders of magnitude. This computational efficiency gain is crucial because to generate each single Constraint (22b) invokes solving $|\mathcal{I}|$ evaluation subproblems (P_S). Please refer to Table 8 in Online Appendix K for more detailed results.

6.1.6. Performance of CG&PS. Next we examine the performance of the CG&PS algorithm for solving the general and convexified instances of (P). Our results show that the CG&PS algorithm solves the general problem (which Gurobi cannot handle) with high precision (i.e., optimality gap $< 10^{-4}$) within seconds even for large instances (more detailed results are documented in Table 9 in Online Appendix K). The efficiency of this algorithm is attributable to two factors. First, only a moderate number of iterations (n_M) are needed, suggesting the merit of iteratively adding constraints to the relaxed master problem. Second, although a large number (n_M) of ESPs (P_S) need to be solved for large instances, the PS algorithm efficiently solves all ESPs with no more than 1.5 seconds.

Furthermore, Table 1 compares the CG&PS algorithm against Gurobi, the BP algorithm, and the customized heuristic algorithm over the convexified problem instances. The table includes the CPU time (in second) of CG&PS (t_{CG}), the number of master problems (MPs) needed to solve (n_M), the average CPU time for solving each MP (t^s_M), the CPU time for solving all MPs (t_M), the number of evaluation subproblems (ESP) needed to solve (n_E), the average CPU time (*in milliseconds*) for solving each ESP (t^s_E), and the CPU time for solving all

Table 1. Performance of the CG&PS Algorithm for Convexified Problems

		CG&PS					Gurobi solver				BP algorithm			Heuristic	
		(P)	MP		ESP		Solved		Unsolved		Root node				
$\lvert\mathcal{I}\rvert$	$\lvert\mathcal{H}\rvert$	t_{CG}(s)	n_M	t^s_M(s)	n_E	t^s_E(ms)	n_G	t_G(s)	gap_1	gap_2	n_P	n_C	gap_3	t_H(s)	gap_4
30	10	0.72	4	0.12	430	0.11	42	410	4.16	0.10	432	31,899	0.18	11.2	0.31
	15	0.53	4	0.08	483	0.12	38	479	2.09	0.17	744	44,253	0.36	6.2	1.92
	20	0.74	5	0.11	601	0.12	37	638	1.61	0.18	141	15,776	0.24	22.1	1.41
60	20	1.20	5	0.15	2,249	0.09	9	475	2.30	0.22	351	131,169	0.55	14.0	3.59
	30	1.10	6	0.12	2,824	0.09	9	908	1.63	0.14	391	120,070	0.55	10.4	3.55
	40	1.58	6	0.16	3,411	0.10	7	1,039	1.76	0.20	148	109,920	1.75	28.9	7.74
90	30	2.24	7	0.21	6,885	0.08	1	1,282	2.38	0.43	284	307,661	1.63	19.7	9.40
	45	2.49	7	0.21	8,441	0.09	—	—	1.52	0.15	302	308,673	1.44	20.6	9.80
	60	3.40	8	0.28	9,299	0.10	—	—	41.16	19.79	162	330,630	3.31	43.6	9.70

ESPs (t_E). In addition, n_G indicates the number of problem instances solved by Gurobi within 1,800 seconds, and t_G indicates the average CPU time (in seconds) for solving these instances. As for instances that Gurobi fails to solve within the time limit, we use gap_1 and gap_2 to measure the quality of solution (z_G, y_G) returned by Gurobi. gap_1 is the relative gap given by Gurobi, that is, $gap_1 = \frac{f(z_G, y_G) - \underline{f}}{\underline{f}}$, where $\underline{f}$ is the lower bound obtained by Gurobi. gap_2 is the gap relative to the optimal objective value f^* given by the CG&PS algorithm, that is, $gap_2 = \frac{f(z_G, y_G) - f^*}{f^*}$. Because the BP algorithm fails to solve the tested problem instances at the root node of the BP tree within 1,800 seconds, we report the number of solved pricing subproblems (n_P) and the number of generated columns (n_C) at the root node. We use $gap_3 = \frac{f^{BP} - f^*}{f^*}$ to measure the performance of the BP algorithm, where f^{BP} is the best incumbent objective value given by the LP relaxation of set-partitioning formulation at the root node within the time limit. If the BP algorithm was able to solve the root node exactly, we should have $gap_3 \leq 0$. For the heuristic algorithm, we report its average CPU time (t_H) and relative gap $gap_4 = \frac{f^H - f^*}{f^*}$, where f^H is the objective value of the solution given by the heuristic algorithm.

Table 1 clearly shows the superior computational efficiency and accuracy of the proposed algorithm. Table 1 shows a speedup of about three orders of magnitude brought by the CG&PS algorithm compared with Gurobi for small instances. As for the large instances, Gurobi fails to solve to optimality within the time limit, yet the CG&PS algorithm spends less than 3.5 seconds before reaching optimality. Table 1 also highlights the necessity of the CG&PS algorithm in providing high-quality solutions because the solutions returned by Gurobi within the time limit or by the heuristic solution may deviate from optimality by a large margin. The BP algorithm is able to produce moderate values of gap_3 within 1,800 seconds. However, the actual quality of its solution may be difficult to assure without knowing f^* because the BP algorithm fails to solve any instance at the root node and the number of columns generated n_C increases dramatically in $|\mathcal{I}|$.

6.2. Managerial Insights

We next discuss the managerial implications of scaling up battery swapping service networks. We modify the numerical setup in Section 6.1 by increasing $|\mathcal{I}|$ and the demand that each swapping station faces (i.e., μ and σ^2) simultaneously and linearly on a scale from 1 to 10. In other words, the total demand increases on a scale from 1 to 100. For the baseline scale, $|\mathcal{I}|$ is set at 10, and μ and σ^2 are at the average level observed from the 10 swapping stations in our data set (as Online Appendix B describes). $|\mathcal{H}|$ is fixed at 35 for all scales. At each scale, we generate 50 problem instances with randomized parameter values and locations of swapping stations and candidate charging stations.

For each problem instance, we solve for Q_i^* and the planning decisions for the main scenario. At the baseline scale, Q_i^* is 17.9 batteries per batch on average, with a standard deviation of 8.9. We also consider three alternative scenarios, namely, the benchmark (i.e., on-site charging) scenario, the benchmark scenario with slow charging speed, and the main scenario yet with $Q_i = 0.3Q_i^*$. To ensure a consistent service level requirement, we set ϵ^B at each swapping station as follows:

$$\epsilon^B = \frac{\mathbb{E}[[[N^S(\Delta) - \frac{1}{n}R]^+ + N^S(T^T) - r]^+]}{Q}, \tag{23}$$

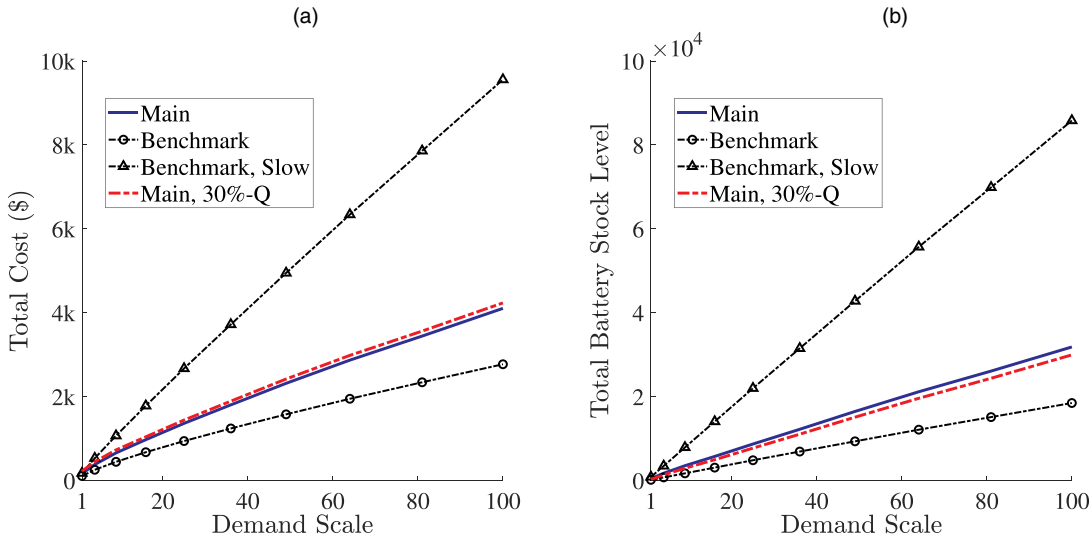
where n is the number of swapping stations that the assigned charging station serves. See Online Appendix L for the justification of (23). Figure 7 illustrates the evolution of the optimal total cost and the optimal total stock level of batteries as the battery swapping service scales up.

6.2.1. Scalability. We first compare the main and benchmark scenarios. As Figure 7 shows, both the total cost and the battery stock level increase (slightly) sublinearly as demand scales up in both scenarios. This economy of scale is owing to the demand pooling effect at each swapping station: As σ^2 increases linearly, the safety stock of batteries of each swapping station increases sublinearly.

However, Figure 7 also shows that the main scenario is less favorable than the benchmark scenario both financially and environmentally, incurring a higher (by 46.92% on average and 48.06% at the 100× scale) cost and a higher (by 93.77% on average and 72.29% at the 100× scale) battery stock level. This is mainly due to the battery asset-heavy nature of centralized charging. That is, all else being equal, centralized charging results in a higher battery stock level by batching battery replenishments and creating transportation lead time.

6.2.2. Urban Grid Capacity and Charge Speed. In light of the previous analysis, the benchmark on-site charging scenario seems preferable toward mass adoption of battery swapping. However, this prospect of on-site charging is dimmed by the urban grid capacity constraint. Most power distribution grids in proximity to urban dwellers are of low voltage (≤ 0.4kV) and were built without allocating enough capacity for mid- or large-scale fast charging with direct current. Grid capacity expansion is expensive and has no longer received major investments in most developed countries. Consequently, distributed charging is expected to largely remain at a slow speed with alternating current. The typical charging power is about 6.2–7.6 kW, as Charge Point specifies.

Figure 7. (Color online) Evolution of (a) Total Optimal Cost and (b) Total Battery Stock Level as Demand Scales Up



If instead assuming the charge power to be as small as 7 kW (which is 17% of the baseline charge speed of 41 kW and the charge time T^C becomes 4.7 hours) at local swapping stations, then Figure 7 shows that the benchmark scenario will incur a higher (by 89.04% on average and 132.82% at the 100× scale) cost and a higher (by 142.03% on average and 169.42% at the 100× scale) battery stock level than the main scenario. This sharp contrast is because the battery stock level generally scales at a rate of $O(\sqrt{T^C})$, or $O\left(\frac{1}{\sqrt{\text{charge speed}}}\right)$. As Figure 8(a) shows, restricting the charge speed will dramatically drive up the battery stock level for both scenarios. Conversely, further enhancing charge speed creates diminishing marginal benefits in terms of saving batteries. This marginal benefit diminishes faster in the main scenario, where the effects of order batching and transportation lead time dominate the effect of the lead time due to battery charging and persist to entail a sizeable battery stock in the system.

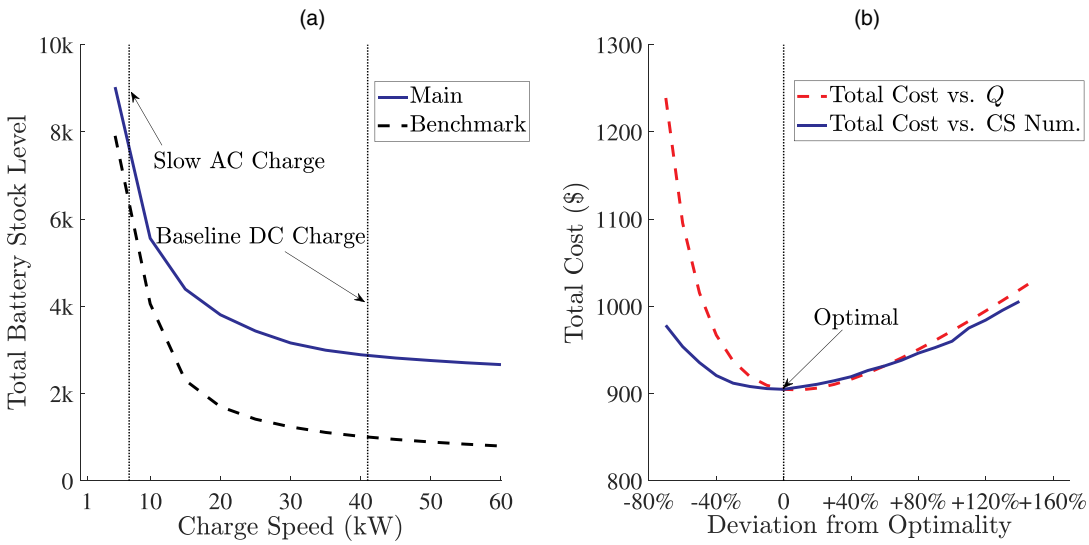
To summarize, the previous results underscore the importance of scaling up battery swapping service capacity in the swap-locally, charge-centrally fashion to ensure sufficient access to grid capacity. Nevertheless, the actual implementation of such systems may have to deviate from the proposed design in the presence of additional complexities, which we investigate as follows.

6.2.3. Flexibility of Reorder Quantity. Will deviating from the optimal reorder quantity Q^* significantly increase the total cost? This question is motivated by the reality where Q^* may not necessarily equal to one truckload. The trucking cost may also be tiered so that only several discrete values of Q are implementable. More importantly, it is environmentally desirable to reduce Q to mitigate the order batching effect and thus reduce the battery stock level.

It turns out that the total cost is a reasonably flat function in Q around its optimal value, as Figure 8(b) illustrates. Such a flat structure is similar to that in the economic order quantity (EOQ) context. As the cost function (19) suggests, the battery asset cost increases in Q, whereas the battery transportation cost decreases in Q. These two effects largely cancel out as Q deviates in either direction from Q^*. Consequently, Figure 7 shows that reducing Q to $0.3Q^*$ can cut the battery stock level by 14.28% yet only increase the cost by 9.92% on average.

6.2.4. Flexibility of Charging Station Density. Finally, urban space often involves multistakeholder complexities, with regulations and permanent structures that may prevent the battery swapping service provider from choosing the optimal locations to deploy charging stations. Fortunately, Figure 8(b) shows that the total cost is also reasonably flat in the number of charging stations that are required to install. Increasing the deployment of charging stations shortens the distance between swapping stations and the central charging station, thus reducing the battery transportation cost and lead time (which reduces the battery asset investment). Conversely, increasing the charging station deployment also incurs a higher charging station building cost. These tradeoffs are balanced around the optimal level. Hence, the service provider can exploit this flexibility to adjust the locations of building charging stations to accommodate the urban complexities that are overlooked in this paper, without incurring significant extra costs.

Figure 8. (Color online) Sensitivity of (a) Total Battery Stock Level to the Charge Speed and (b) Total Cost to the Deviations of the Order Quantities and Number of Charging Stations from the Optimal Solution



Note. The number of swapping stations and demand are set as five times the baseline values.

7. Conclusion

This paper is motivated by the looming tensions in electrifying urban mobility: the upscaling needs for battery swapping versus the mounting challenges in ensuring battery proximity and availability for customers and grid accessibility for charging operations. To understand and then address these tensions, we examine a swap-locally, charge-centrally network configuration. We develop data-motivated models of battery circulation and citywide infrastructure deployment. The analysis of battery circulation enriches the classical repairable-inventory theory. We also make methodological contributions of efficiently solving a new location-inventory problem which is high-dimensional, nonconvex, and nonconcave. The proposed algorithmic framework extends the state-of-art scope of the problems that can be solved by using submodularity. Our models lead to insights. (1) Centralizing battery charging is less (or more) scalable than the status-quo on-site charging at swapping stations, if on-site charging has (or has no) access to grids that permit fast charging. (2) If centralizing charging, the service providers can exploit additional flexibilities in adjusting their design and operations to accommodate additional urban complexities.

This paper analytically models and understands scaling up battery swapping services in our cities in the coming decades. More research questions pertaining to this emerging paradigm are awaiting cross-disciplinary (e.g., management science, transportation, and power systems) efforts to answer. For example, it will be beneficial to study alternative network configurations, such as battery lateral transshipment among charging or swapping stations. Moreover, centralizing charging may allow the operator to participate in the energy and capacity markets of electricity, whereas decentralizing charging may provide local microgrids with additional reserve. Analyzing such tradeoffs will deepen our understanding of mobility-energy orchestration toward a smart-city future.

Acknowledgments

The authors thank Chung-Piaw Teo (department editor), the associate editor, and two anonymous referees for valuable review comments. The authors also thank the National Monitoring and Management Center for New Energy Vehicles in China for providing data. The first two authors share co-first authorship.

References

Atamtürk A, Gómez A (2020) Submodularity in conic quadratic mixed 0–1 optimization. *Oper. Res.* 68(2):609–630.

Atamtürk A, Narayanan V (2008) Polymatroids and mean-risk minimization in discrete optimization. *Oper. Res. Lett.* 36(5):618–622.

Atamtürk A, Berenguer G, Shen ZJM (2012) A conic integer programming approach to stochastic joint location-inventory problems. *Oper. Res.* 60(2):366–381.

Avci B, Girotra K, Netessine S (2015) Electric vehicles with a battery switching station: Adoption and environmental impact. *Management Sci.* 61(4):772–794.

Axsäter S (2015) *Inventory Control*, vol. 225 (Springer, Berlin).

Bloomberg (2019) The best way to boost electric cars in China. Accessed May 1, 2020, https://www.bnnbloomberg.ca/the-best-way-to-boost-electric-cars-in-china-1.1270493.

BNEF (2019) Electric vehicle outlook 2019. Accessed May 1, 2020, https://about.bnef.com/electric-vehicle-outlook/.

Electrive (2021) NIO presents next-gen battery swap station with sinopec. Accessed June 1, 2021, https://www.electrive.com/2021/04/15/nio-presents-next-gen-battery-swap-station-with-sinopec.

Fischetti M, Ljubić I, Sinnl M (2017) Redesigning benders decomposition for large-scale facility location. *Management Sci.* 63(7): 2146–2162.

Graves SC (1985) A multi-echelon inventory model for a repairable item with one-for-one replenishment. *Management Sci.* 31(10): 1247–1256.

Greenblatt JB, Saxena S (2015) Autonomous taxis could greatly reduce greenhouse-gas emissions of US light-duty vehicles. *Nature Climate Change* 5(9):860–863.

Guide VDR Jr, Srivastava R (1997) Repairable inventory theory: Models and applications. *Eur. J. Oper. Res.* 102(1):1–20.

Hasija S, Shen ZJM, Teo CP (2020) Smart city operations: Modeling challenges and opportunities. *Manufacturing Service Oper. Management* 22(1):203–213.

IEA (2020) Tracking transport. Accessed June 1, 2021, https://www.iea.org/reports/tracking-transport-2020.

Lee HL, Moinzadeh K (1987) Two-parameter approximations for multi-echelon repairable inventory models with batch ordering policy. *IIE Trans.* 19(2):140–149.

Li Y, Shu J, Wang X, Xiu NH, Xu DC, Zhang JW (2013) Approximation algorithms for integrated distribution network design problems. *INFORMS J. Comput.* 25(3):572–584.

Liu XC, Zhao TY, Yao SH, Soh CB, Wang P (2019) Distributed operation management of battery swapping-charging systems. *IEEE Trans. Smart Grid* 10(5):5320–5333.

Mak HY (2022) Enabling smarter cities with operations management. *Manufacturing Service Oper. Management* 24(1):24–39.

Mak HY, Rong Y, Shen ZJM (2013) Infrastructure planning for electric vehicles with battery swapping. *Management Sci.* 59(7):1557–1575.

Moinzadeh K, Lee HL (1986) Batch size and stocking levels in multi-echelon repairable systems. *Management Sci.* 32(12):1567–1581.

Muratori M (2018) Impact of uncoordinated plug-in electric vehicle charging on residential power demand. *Natural Energy* 3(3): 193–201.

NBC (2020) Britain to ban the sale of new gas, diesel and hybrid cars from 2035. Accessed May 1, 2020, https://www.nbcnews.com/news/world/britain-ban-sale-new-gas-diesel-hybrid-cars-2035-n1129616.

Ni W, Shu J, Song M, Xu D, Zhang K (2021) A branch-and-price algorithm for facility location with general facility cost functions. *INFORMS J. Comput.* 33(1):86–104.

Pak-Poy P (1964) The use and limitation of the poisson distribution in road traffic. *Proc. 2nd Australian Road Res. Board Conf.* (TRB, Washington, DC), 223–247.

Perera S, Janakiraman G, Niu SC (2018) Optimality of (s, s) inventory policies under renewal demand and general cost structures. *Production Oper. Management* 27(2):368–383.

Qi W, Shen ZJM (2019) A smart-city scope of operations management. *Production Oper. Management* 28(2):393–406.

Shanker K (1981) Exact analysis of a two-echelon inventory system for recoverable items under batch inspection policy. *Naval Res. Logist.* 28(4):579–601.

Shen ZJM, Qi L (2007) Incorporating inventory and routing costs in strategic location models. *Eur. J. Oper. Res.* 179(2):372–389.

Shen ZJM, Coullard C, Daskin MS (2003) A joint location-inventory model. *Transportation Sci.* 37(1):40–55.

Sherbrooke CC (1968) Metric: A multi-echelon technique for recoverable item control. *Oper. Res.* 16(1):122–141.

Sherbrooke CC (2006) *Optimal Inventory Modeling of Systems: Multi-Echelon Techniques*, vol. 72 (Springer Science & Business Media, Boston).

Shu J (2010) An efficient greedy heuristic for warehouse-retailer network design optimization. *Transportation Sci.* 44(2):183–192.

Shu J, Teo CP, Shen ZJM (2005) Stochastic transportation-inventory network design problem. *Oper. Res.* 53(1):48–60.

Simon RM (1971) Stationary properties of a two-echelon inventory model for low demand items. *Oper. Res.* 19(3):761–773.

Sivazlian BD (1974) A continous-review (s, S) inventory system with arbitrary interarrival distribution between unit demand. *Oper. Res.* 22(1):65–71.

Snyder LV, Daskin MS, Teo CP (2007) The stochastic location model with risk pooling. *Eur. J. Oper. Res.* 179(3):1221–1238.

Svoronos A, Zipkin P (1988) Estimating the performance of multi-level inventory systems. *Oper. Res.* 36(1):57–72.

Teo CP, Shu J (2004) Warehouse-retailer network design problem. *Oper. Res.* 52(3):396–408.

Wu OQ, Yücel S, Zhou YF (2021) Smart charging of electric vehicles: An innovative business model for utility firms. *Manufacturing Service Oper. Management* 24(5):2481–2499.

Xu XH, Yao LZ, Zeng PL, Liu YJ, Cai TT (2015) Architecture and performance analysis of a smart battery charging and swapping operation service network for electric vehicles in China. *J. Modelling Power Systems Clean Energy* 3(2):259–268.

Zhang X, Wang GB (2016) Optimal dispatch of electric vehicle batteries between battery swapping stations and charging stations. *Proc. IEEE Power and Energy Society General Meeting*, 1–5.

Zhang YL, Shen ZJM, Song SJ (2016a) Distributionally robust optimization of two-stage lot-sizing problems. *Production Oper. Management* 25(12):2116–2131.

Zhang YL, Shen ZJM, Song SJ (2016b) Parametric search for the bi-attribute concave shortest path problem. *Transportation Res. Part B: Methodological* 94:150–168.

Zhang YL, Shen ZJM, Song SJ (2018) Exact algorithms for distributionally β-robust machine scheduling with uncertain processing times. *INFORMS J. Comput.* 30(4):662–676.

Copyright 2023, by INFORMS, all rights reserved. Copyright of Management Science is the property of INFORMS: Institute for Operations Research and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.